

AI-Based Computer Vision and Deep Learning Approach for Multiple Sclerosis (MS) Lesion Detection

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FINAL YEAR DESIGN PROJECT REPORT

**This Report Presented in Partial Fulfillment of the
Requirements for the Degree of Bachelor of Science in
Computer Science and Engineering**

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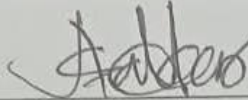
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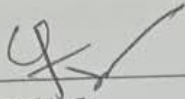
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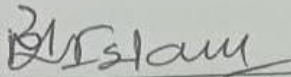
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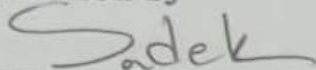


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We hereby declare that this project has been done by us under the supervision of **Mr. Md. Sadekur Rahman**, Assistant Professor, Department of Computer Science and Engineering, Daffodil International University. We also declare that neither this project nor any part of this project has been submitted elsewhere for the award of any degree or diploma.

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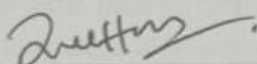


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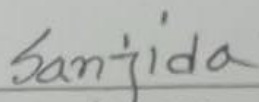
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ABSTRACT

Multiple Sclerosis (MS) is a chronic neurological disorder characterized by the formation of demyelinating lesions in the central nervous system, which can be identified through magnetic resonance imaging (MRI). Accurate segmentation of MS lesions is essential for supporting disease assessment and quantitative analysis; however, automated lesion segmentation remains challenging because of the small size, irregular morphology, low contrast, and substantial variability of lesions across MRI scans. This study presents a deep learning-based framework for automated MS lesion segmentation using multimodal three dimensional MRI data, with particular emphasis on preserving the discriminative information available in FLAIR while exploiting complementary information from T1- and T2-weighted images. The proposed framework employs N4 bias-field correction followed by non-zero voxel Z-score normalization, while preserving the background intensity as zero. A FLAIR-only UNETR model was first established as a baseline using the preprocessed MRI volumes. Subsequently, a custom three-dimensional gated residual multimodal architecture was developed in Experiment 7 Phase 2, where FLAIR is used as the primary branch and T1 and T2 are incorporated through an auxiliary branch. The multimodal information is integrated through gated feature fusion and a bounded residual correction mechanism, allowing the auxiliary modalities to refine the FLAIR-based segmentation without replacing its primary representation. The experiments were conducted on the MSLesSeg dataset using patient-level evaluation, with a held-out test set containing 22 patients and 874 annotated lesions. The proposed model achieved a global Dice coefficient of 70.09%, global precision of 66.54%, global recall of 0.7406, and global IoU of 53.96%, while achieving a patient-wise mean Dice of 64.00%. At the lesion level, mean recall improved from 42.00% with the FLAIR-only baseline to 48.30% with the proposed multimodal model, while lesion detection rate increased from 66.13% to 71.62%. The analysis further demonstrated that the proposed approach provided the largest recall improvements for several low-brightness and large-lesion groups, indicating improved sensitivity to challenging lesions. Finally, a user oriented MRI segmentation interface was developed to demonstrate the practical use of the trained models for FLAIR-only and multimodal inputs. Overall, the results demonstrate the feasibility of a FLAIR-preserving gated multimodal deep learning framework for automated MS lesion segmentation and highlight the potential of complementary MRI modalities for improving lesion detection and segmentation performance.

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Chapter 1

Introduction

This chapter introduces the research on automated Multiple Sclerosis (MS) lesion segmentation from brain MRI using deep learning and multimodal image analysis. It begins by discussing the importance of accurate lesion segmentation and the difficulties involved in manual lesion delineation, particularly when lesions differ in size, shape, intensity, and appearance. The chapter then presents the motivation for using deep learning with three-dimensional MRI data and describes the overall direction of the proposed work. The FLAIR-only UNETR model is introduced as the baseline, followed by the proposed gated residual multimodal architecture, which combines FLAIR, T1-weighted, and T2-weighted MRI. The chapter also outlines the research objectives, methodology, expected outcomes, and overall organization of the report.

1.1 Introduction

Multiple Sclerosis (MS) is a chronic neurological disorder that affects the central nervous system and can lead to the development of lesions in the brain and spinal cord. Magnetic Resonance Imaging (MRI) is widely used to assess MS because it provides detailed information about brain tissue and helps identify and monitor these lesions. Accurate segmentation of MS lesions is important for quantitative analysis, disease assessment, treatment monitoring, and research. However, identifying the lesions precisely is difficult because they can be small, irregular in shape, have low contrast, and vary considerably in appearance. In conventional practice, lesion segmentation often depends on manual delineation by clinical experts. Although this can produce reliable annotations, the process requires considerable time and effort and may vary between observers. As the use of MRI in MS assessment increases, relying entirely on manual segmentation becomes less practical. These limitations have increased the need for automated methods that can assist clinicians and researchers with lesion identification in a more efficient and consistent manner.

Deep learning has become increasingly useful in medical image analysis because of its ability to learn relevant features directly from imaging data. Convolutional Neural Networks (CNNs) are effective at learning local spatial and textural features, while Transformer-based models can capture relationships across larger regions of an image. UNETR, for example, applies Transformer-based feature representation to three-dimensional medical image segmentation. Even with these developments, MS lesion segmentation remains difficult because lesions differ in size, intensity, location, and morphology. Relying on a single MRI sequence may also limit the information available for identifying some lesions. Different MRI modalities can provide complementary information: FLAIR is particularly useful for making MS lesions more visible, while T1- and T2-weighted images provide additional structural and intensity information. However, simply combining these modalities does not necessarily mean that the network will use each one effectively. For this reason, this research investigates a selective multimodal strategy that keeps FLAIR as the primary source of lesion information while using T1 and T2 to provide additional information when useful.

The proposed framework applies N4 bias-field correction followed by non-zero voxel Z-score normalization before model processing. A FLAIR-only UNETR model is first used as the baseline. The proposed Experiment 7 Phase 2 model then extends this setup by using separate branches for the MRI modalities. FLAIR is processed through the primary branch, while T1 and T2 are handled by an auxiliary branch. Their features are combined using gated feature fusion, followed by bounded residual correction. In this way, the additional modalities are used to refine the FLAIR-based representation rather than completely replacing it.

The performance of the models is evaluated using quantitative segmentation metrics together with lesion-level analysis. This enables the study to analyze both the quality of the segmentation and the detection of individual lesions under various conditions by the models. The system also offers a user friendly interface to display the MRI data and the segmentation results. These components assess the ability of controlled multimodal feature integration to boost MS lesion detection and segmentation without sacrificing the valuable lesion information offered by FLAIR.

1.2 Motivation

The main motivation for this research is the need for an accurate and efficient way to segment Multiple Sclerosis (MS) lesions from brain MRI. Manual lesion delineation is time consuming and depends on clinical expertise. Although MRI is commonly used to examine abnormalities related to MS, it can be difficult to manually identify and outline each lesion and the results may differ from one observer to another. This workload can be reduced by an automated deep learning-based method that can provide consistent lesion localization and segmentation, which can also facilitate quantitative analysis by clinicians and researchers. Another major factor is the difference in the areas of lesions of MS. The size, shape, intensity, location and overall appearance of the lesions can vary significantly. In particular, small lesions and lesions with low intensity can be difficult to distinguish from the surrounding brain tissue. A model may therefore obtain a good overall segmentation score while still missing some individual lesions. For this reason, this research gives additional attention to lesion-level detection and sensitivity instead of relying only on an overall segmentation metric.

The availability of multiple MRI modalities also motivates this study. FLAIR is useful for making MS lesions more visible, while T1- and T2-weighted images contain additional structural and intensity information. These differences suggest that the modalities can complement one another during segmentation. However, combining them directly does not necessarily ensure that the model will make effective use of the information from each sequence. This led to the use of a selective fusion strategy in which FLAIR remains the primary modality, while T1 and T2 provide additional information to refine the segmentation.

The proposed gated residual learning strategy is another important part of the motivation. In this approach, FLAIR is treated as the main source of lesion-related information, whereas T1 and T2 are used as auxiliary inputs. The gated feature fusion mechanism controls how much information from the auxiliary branch is incorporated. After fusion, the bounded residual correction provides a controlled way for the multimodal information to refine the FLAIR-based prediction. The intention is to gain useful information from the additional MRI sequences without losing the lesion representation already provided by FLAIR.

Finally, this research considers that overall segmentation metrics alone may not provide a complete picture of model performance. Therefore, precision, recall, Dice coefficient, and IoU are evaluated together with lesion-level detection and recall. The analysis also examines performance according to lesion size and FLAIR brightness to identify which types of lesions remain difficult for the models. A user-oriented MRI segmentation interface is included as well, providing a practical way to present the trained models and their segmentation results for future research and potential clinical-support use.

1.3 Objectives

- i. To devise and test a FLAIR based deep learning model for automated MS lesion segmentation and to create a UNETR based baseline model for 3D MRI segmentation.
- ii. To design and implement a gated residual multimodal architecture that takes the FLAIR modality as the primary modality and adds complementary information from T1- and T2-weighted MRI via gated feature fusion and bounded residual correction.
- iii. To measure and compare the performance of the baseline and proposed models in terms of Precision, Recall, Dice Coefficient and Intersection over Union (IoU). Further lesion-level analysis will be conducted to assess the models' performance on tough-to-find MS lesions.
- iv. To investigate the influence of the lesion characteristics on the segmentation performance by analyzing the model behavior on various sizes of lesions and FLAIR brightness levels.

1.4 Methodology

The proposed methodology follows a structured pipeline for automated three-dimensional Multiple Sclerosis (MS) lesion segmentation from multimodal brain MRI scans. The process starts with the MSLesSeg dataset, which contains FLAIR, T1-weighted, T2-weighted, and corresponding lesion mask volumes. Before model training, the MRI volumes undergo a two-stage intensity preprocessing procedure. N4 bias-field correction is first applied to the skull-stripped MRI volumes using a foreground mask derived from non-zero voxels. Non-zero voxel Z-score normalization is then performed, with the mean and standard deviation calculated only from the non-zero regions while the zero-valued background remains unchanged.

After preprocessing, two model configurations are investigated. The first is a FLAIR-only UNETR model, which serves as the baseline for three-dimensional lesion segmentation. The second is the proposed Experiment 7 Phase 2 architecture, where FLAIR is processed through a primary branch and T1- and T2-weighted images are processed through an auxiliary branch. Features from these branches are combined using gated feature fusion, and a bounded residual correction mechanism is subsequently used to refine the FLAIR based prediction with additional multimodal information.

The models are trained using three-dimensional MRI data, and sliding-window inference is applied to generate predictions for the complete MRI volumes. Precision, Recall, Dice Coefficient, and Intersection over Union (IoU) are used to evaluate their performance.

Besides the segmentation performance, lesion-level analysis is performed to evaluate the detection and recall of each lesion. In addition, lesion size and FLAIR brightness are also taken into account, to study the performance of the models on different and possibly difficult groups of lesions.

1.5 Project Outcome

The ultimate goal of this work is to develop a fully automatic Multiple Sclerosis (MS) lesion segmentation framework based on deep learning from 3D brain MRI volumes. The goal of the project is to show how FLAIR could be the main imaging modality and how additional information from T1- and T2-weighted MRI can be added in a gated residual multimodal architecture. By comparing the FLAIR-only UNETR baseline to the proposed multimodal approach, the benefits of incorporating other MRI modalities for MS lesion segmentation can be explored.

The project also assesses the segmentation performance at the overall level and at the level of lesion. Lesion size and FLAIR brightness is given special consideration to better understand the model performance in challenging, small and low-contrast lesions. These analyses can be helpful to identify strengths and limitations of multimodal segmentation and can help guide future studies aimed at improving automated MS lesion analysis.

Lastly, the framework created could be useful for researchers and clinicians by decreasing the manual workload of delineating lesions and improving the quantitative analysis of MRI data in a more consistent manner. The project thus serves to the application of deep learning and computer vision for medical image analysis with a special emphasis on automated MS lesion detection and segmentation.

1.6 Organization of the Report

This report is organized into six chapters, covering the main stages of the research, system development, implementation, and evaluation process:

- i. Chapter 1 Introduction: Introduces Multiple Sclerosis, MRI-based lesion segmentation, and the motivation behind the research. It also outlines the research objectives, methodology, expected outcomes, and overall scope of the project.
- ii. Chapter 2 Background: Discusses the fundamental concepts related to MS lesion segmentation, medical image analysis, deep learning, CNN- and Transformer-based segmentation methods, multimodal MRI analysis, and relevant existing research. The chapter also identifies the research gaps addressed by the proposed approach.
- iii. Chapter 3 Research Methodology: Describes the MSLesSeg dataset and the complete methodology used in the project. It covers the MRI preprocessing pipeline, including N4 bias-field correction and non-zero Z-score normalization, the FLAIR-only UNETR baseline, the proposed gated residual multimodal architecture, training strategy, system design, and evaluation procedures.
- iv. Chapter 4 Implementation and Results: Presents the hardware and software environment, experimental configurations, model training and evaluation procedures,

and the comparative results of the evaluated models. It also provides detailed analysis of overall segmentation performance and lesion-level results.

v. Chapter 5 Engineering Standards and Design Challenges: Discusses the software, hardware, and communication standards considered during development, along with ethical aspects of medical imaging and AI-assisted analysis. The chapter also covers computational limitations, societal and environmental impacts, sustainability considerations, project management, financial analysis, and the complex engineering challenges encountered during the project.

vi. Chapter 6 Conclusion: Summarizes the major findings and contributions of the research, discusses the limitations of the developed approach, and presents possible directions for future work, including improved multimodal learning, lesion-level detection, and further validation.

Chapter 2

Background

This chapter provides the background needed to understand automated Multiple Sclerosis (MS) lesion segmentation from brain MRI. It introduces the clinical and technical challenges involved in identifying MS lesions and discusses the use of deep learning for medical image segmentation. The chapter also reviews CNN- and Transformer-based segmentation approaches, multimodal MRI analysis, and existing methods for automated MS lesion segmentation. Finally, the limitations of current approaches are discussed to establish the research gap addressed by the proposed FLAIR-preserving gated residual architecture.

2.1 Introduction

Multiple Sclerosis (MS) is a chronic neurological disease that affects the central nervous system and is associated with inflammatory and demyelinating processes that can lead to lesion formation in the brain and spinal cord. Magnetic Resonance Imaging (MRI) plays an important role in identifying and monitoring MS lesions, while accurate lesion segmentation can support quantitative assessment, disease monitoring, and clinical research. Automated MS lesion segmentation has developed from conventional image-processing and machine-learning techniques toward end-to-end deep learning approaches. Convolutional Neural Networks (CNNs) are effective at learning local spatial and textural features, whereas Transformer-based architectures can capture relationships across larger regions of three-dimensional medical images. In parallel, multimodal MRI segmentation has investigated the complementary information provided by FLAIR, T1-weighted, and T2-weighted images.

However, there are still a number of difficulties with MS lesion segmentation, as lesions can vary significantly in size, shape, intensity and spatial distribution. With small and low contrast lesions, it is especially hard to find them accurately. Another challenge is the effective fusion of multiple MRI modalities, as just a simple concatenation of features may not be sufficient to control the extent of information provided by different modalities. This chapter summarizes the related studies on automated MS lesion segmentation, examines the current methods and related applications, and outlines the research gaps which drive the proposed methodology.

2.2 Literature Review

Over the years, studies of automated Multiple Sclerosis (MS) lesion segmentation have transitioned from traditional image-processing and machine-learning methods to more sophisticated deep learning approaches. Previous approaches relied primarily on hand-crafted intensity, texture and spatial features. However, due to the emergence of Convolutional Neural Networks (CNNs), lesion features might be learned directly from the MRI data using end-to-end training. Recent studies have investigated various methods such as multi-branch networks, attention based models, recurrent modules, transfer learning, Transformer based encoders, self ensembling, diffusion models, and federated learning. However, the detection of small, sparse, irregular and low-contrast

MS lesions is still challenging. Variations in performance may also occur if MRI acquisition protocols, scanners or patient populations vary.

To enhance the in-domain performance as well as out-of-domain generalization, Zhang et al. [1] proposed a unified MS lesion segmentation framework named UNISELF. To mitigate domain shifts and the lack of MRI contrasts, the framework included instance normalization and test-time self ensembled lesion fusion. It had a Dice score of around 0.664, as well as a PPV of 0.882 and used extra evaluations on external data sets. It is important that this work addresses two important practical problems in multimodal segmentation in real MRI data: the first being the difficulty of handling multimodal MRI data from different imaging domains, and the second being the lack of specific MRI contrasts.

Hindawi et al. [2] studied federated learning for the purpose of MS lesion segmentation in several clinical centers. They've applied a self-configuring nnU-Net in a federated learning paradigm, where the models can be learned without sensitive clinical images being explicitly shared among institutions. The cross-site ensemble that was obtained yielded Dice scores of 0.66 to 0.80 on held-out sets of images. The study not only illustrates the power of privacy-preserving distributed learning, but also highlights the impact of domain variation and scanner differences on segmentation performance.

Andishgar et al. [3] introduced a three-dimensional U-Net model (R2AUNet) using a recurrent residual model and dual attention gates for FLAIR-only MS lesion segmentation. It has been trained with 112 MRI images of 95 patients and it obtained a dice similarity coefficient of 0.768, a precision of 0.825 and a sensitivity of 0.765 on the test set. The authors reported that it could enhance boundary refinement of lesions and detection of small and low-contrast lesions with the architecture. This study is especially relevant as it demonstrates that the well-designed FLAIR-only architecture can perform well in the evaluation setting of this study, and indicates the potential benefit of residual and attention-based feature learning.

Gaj et al. [4] explored subject-based transfer learning for the segmentation of MS lesions longitudinally. They first used the first scan of each subject to adapt a pretrained model and then applied the adapted representation to the later scans. On ISBI 2015, the method obtained a DSC of 79.2%. The results show that patient-specific information can be used to improve longitudinal segmentation, but the method requires previous scans, and thus cannot be directly applied to newly presented patients.

Belwal and Singh [5] tested the modified U-Net for segmentation of MS lesions and achieved a Dice score of 73.98% on the test set, precision between 73-74% and recall between 91-94%. Results indicate that a suitable modification and training of the encoder-decoder can still achieve competitive results in segmentation. Meanwhile, training-testing performance gap highlights the need to assess generalization and not just training performance.

The MSLesSeg dataset, presented by Guarnera et al. [6] in 2025, is another significant advancement in the field of MS lesion segmentation. The data set includes 115 scans of 75 patients with T1-weighted, T2-weighted and FLAIR sequences and expert validated lesion annotation. It has been created to facilitate research in lesion segmentation for MS and serve as a standard for the evaluation of automated methods in a multimodal scenario. Its application of several MRI sequences is especially pertinent to research studies that aim to determine whether other modalities can add value to FLAIR imaging.

The MSLesSeg benchmark is also very helpful for models based on transformers for segmentation. In the dataset paper, the mean Dice score of MSSegDiff is 0.6851, while the mean scores of UNETR and SwinUNETR are 0.6421 and 0.6786 respectively and the mean score of TransBTS is 0.4917. The individual UNETR fold Dice scores were between 0.6347 and 0.6541. The results of this experiment show a significant difference in performance for Transformer-based architectures across the datasets and the evaluation protocol. They also show that it is not always the case that overall Dice and lesion sensitivity are measures of the same quality of segmentation. The MSLesSeg study also gives interesting insights into multimodal MRI. The FLAIR sequence was determined to be useful for the detection of lesions, and T1- and T2-weighted images were found to be complementary. Multimodal analysis: each MRI sequence provides a different tissue contrast which can be useful in the analysis of MS lesions. The consensus analysis also highlighted that segmentation errors may be higher when only data from the FLAIR is available, for lesions that are larger and/or in unusual locations at the boundaries of the lesions. The observations of these cases motivate the study of controlled multimodal feature integration.

Zhu et al. [7] evaluated four models, YOLOv9e, YOLOv9c, UNeXt and nnU-Net on the MICCAI 2016 MS lesion segmentation challenge dataset. YOLO based methods yielded Dice scores of ~ 0.57 , UNeXt and nnU-Net yielded Dice scores of 0.79 and 0.76, respectively. This comparison indicates that architectures that are developed specifically for dense segmentation may be more appropriate for detailed lesion boundary recovery than architectures developed mainly for object localization. It also focuses on the need to choose an architecture based on the nature of volumetric lesion segmentation, and not as a standard object detection problem.

Pishvai et al. [8] studied brain parcellation in the context of a diffusion based framework for segmentation of MS lesions. The study achieved a Dice score of 74.14% on the dataset of ISBI 2015. While the overall segmentation accuracy averaged across the entire brain did not significantly change with the brain parcellation, the brain parcellation did enhance the robustness of the segmentation across the different anatomical regions and enhanced the sensitivity to small lesions. This indicates that anatomical information could be useful when it comes to dealing with regional variation and difficult lesion cases, despite modest gains in overall segmentation performance.

In 2024, Wiltgen et al. [9] introduced LST-AI, a deep learning ensemble for automated MS lesion segmentation. The system has been trained with 491 pairs of T1-weighted and FLAIR images and tested on 103 publicly available test cases. It achieved a Dice score of 0.65 and an F1 score of 0.63 on the MSSEG-1 challenge set and overall scores of over 0.62 on Dice and F1. The study also revealed that the detection rate of lesions increased with lesion volume, suggesting that the detection rate was lower for smaller lesions. This work is of particular relevance to the present research as it demonstrates that even the ensemble-based segmentation system can have limitations in terms of sensitivity at the lesion level. It also highlights the importance of analyzing model performance based on individual lesion characteristics, as opposed to an overall segmentation score only.

Thabet et al. [10] in their review work have presented a detailed overview of the machine learning and deep learning approaches used for segmentation and classification of MS lesions from MRI neuroimaging modalities from the year 2019 to 2023. The review systematically categorized the current approaches into two groups, namely segmentation and classification, each of which was subdivided in the sense of supervised and

unsupervised learning strategies. Architectures based on U-Net achieved the best results and were the most prevalent for segmentation, with the highest reported DSC of 90% from a multimodal U-Net trained on FLAIR, PD-w, T1-w, and T2-w sequences. In terms of classification, all modern DL techniques outperformed classical ML techniques in all performance measures. In addition, the review noted several longstanding issues in the field, such as the lack of publicly available labeled datasets, the class imbalance between lesion and non-lesion areas, the lack of detection of small lesions, and the inability to obtain multi-modal MRI sequences in clinical practice.

At the IEEE International Conference on Bioinformatics and Biomedicine 2023, Rondinella et al. [11] introduced a diffusion-based approach to segmenting MS lesions from multimodal MRI. The work shows that generative and diffusion based methods are increasingly being used for medical image segmentation and that more and more multi-modal information is now being used in MS lesion analysis.

Wahlig et al. [12] analysed the transfer learning approach to MS lesion segmentation using a private dataset of 149 patients. The study took into account a number of segmentation tasks, such as segmentation of FLAIR lesions at a single time point, new lesion detection from longitudinal scans, and improving lesion segmentation with post-contrast T1 images. For this enhancing lesion segmentation task, the transfer learning approach significantly outperforms training from scratch with lesion-wise sensitivity of 0.74 vs. 0.44 and PPV of 0.69 vs. 0.37. The findings indicate that, in the case of small amounts of labelled medical data, some pretrained representation can be beneficial, but adaptation to the subjects and the segmentation task might still be required.

Aligned with this goal, Krishnan et al. [13] created a three-dimensional multi-arm U-Net to segment T2 lesions from data acquired from relapsing MS clinical trials. The mean internal DSC was 0.66 or higher and the mean external DSC was 0.62 or higher, and the mean sensitivity was 0.72 or higher and 0.68 or higher, respectively, for the method. The study also found that the performance of the models was reduced when dealing with smaller lesions. This observation is significant when considering MS lesion segmentation, given the challenges of detecting small lesions, and in advocating the inclusion of lesion-level evaluation alongside aggregate segmentation metrics.

Sarica et al. [14] proposed a Dense-Residual U-Net with Attention Gates, Efficient Channel Attention, and Atrous Spatial Pyramid Pooling. The three-plane 2D fusion strategy used by this design scored 0.6688 mean dice score on ISBI 2015. Dense and residual blocks were employed to enhance the propagation of information and the attention mechanisms were applied to improve the attention of the network to the relevant lesion area and suppress the response from the irrelevant background area. In this paper, we demonstrate the progress of U-Net based architectures by the incorporation of residual learning, attention and multi-scale feature extraction.

Overall, the reviewed studies show a gradual shift from conventional U-Net and CNN architectures toward multi-branch fusion, lesion-aware learning, attention mechanisms, recurrent contextual modeling, transfer learning, Transformer-based encoders, self-ensembling, diffusion models, and federated learning. Although several studies have achieved strong results within their respective experimental settings, recurring challenges include the detection of small and low-contrast lesions, scanner and domain variation, limited annotated datasets, missing MRI modalities, and the effective integration of complementary information from multiple MRI sequences.

An important point when comparing these reported results is that performance depends strongly on the evaluation protocol. For instance, LST-AI achieved a Dice score of 0.65 on MSSEG-1, while UNISELF's ISBI-based evaluation yielded a score around 0.664 and R2AUNet achieved a score of 0.768 on its own three-dimensional FLAIR test dataset. The values presented cannot be directly compared to each other since the datasets, patient groups, preprocessing procedures, and experimental conditions vary accordingly. Rather, they suggest that none of the architectural solutions are consistently best for the different MS lesion segmentation configurations.

From the literature reviewed, one of the key research areas is selective multimodal feature integration. Previous studies have shown that other MRI sequences can be of value as supplementary data, and that FLAIR is very informative in the identification of MS lesions. The challenge is, therefore, not only to integrate information from the various modalities, but to regulate the timing and manner in which information from the auxiliary modalities influences the primary lesion representation. This consideration inspires the approach taken in the present research, which includes an auxiliary branch with T1 and T2, gated feature fusion and bounded residual correction. Experiment 6 is used to set a baseline FLAIR-only UNETR, and Experiment 7 Phase 2 investigates the effectiveness of controlled multimodal refinement on both overall segmentation and lesion-level detection.

Table 2.1: Summary of Literature Reviewed.

Author (s)	Year	Title	Methodology	Key Findings
Zhang et al. [1]	2026	UNISELF: A Unified Network with Instance Normalization and Self-Ensembled Lesion Fusion for Multiple Sclerosis Lesion Segmentation	Unified U-Net-based architecture using instance normalization and test-time self-ensembled lesion fusion.	Improved segmentation accuracy and generalization across different datasets, scanners, and imaging conditions.
Hindawi et al. [2]	2025	Federated Learning for Lesion Segmentation in Multiple Sclerosis: A Real-World Multi-Center Feasibility Study	Self-configuring nnU-Net combined with federated learning across multiple clinical centers without sharing raw patient data.	Achieved Dice scores of approximately 0.66–0.80 across sites, demonstrating the potential of privacy-preserving multi-center learning.
Andishgar et al. [3]	2025	Automated Multiple Sclerosis Lesion Segmentation from 3D-FLAIR MRI Using R2AUNet: A Deep Learning Approach with Recurrent Residual and Attention Mechanisms	3D FLAIR-based R2AUNet incorporating recurrent residual blocks and dual attention mechanisms.	Achieved DSC of 0.768, precision of 0.825, and sensitivity of 0.765, showing strong FLAIR-only segmentation performance.

Gaj et al. [4]	2025	Subject-Based Transfer Learning in Longitudinal Multiple Sclerosis Lesion Segmentation	Subject-specific transfer learning using an earlier MRI scan to adapt the model to subsequent longitudinal scans.	Achieved approximately 79.2% DSC on the ISBI 2015 dataset and showed the potential of subject-specific adaptation.
Belwal & Singh [5]	2025	Multiple Sclerosis Lesion Segmentation Using MRI: U-Net Approach	Modified U-Net evaluated on the ISBI 2015 dataset.	Achieved 73.98% test Dice with high recall, demonstrating effective lesion segmentation.
Guarnera et al. [6]	2025	MSLesSeg: Baseline and Benchmarking of a New Multiple Sclerosis Lesion Segmentation Dataset	Introduced the multimodal MSLesSeg dataset with T1, T2, and FLAIR MRI and expert lesion annotations; benchmarked MSSegDiff, UNETR, SwinUNETR, and TransBTS.	Five-fold mean Dice scores were 0.6851, 0.6421, 0.6786, and 0.4917, respectively. The study highlighted the importance of FLAIR and complementary information from T1/T2.
Zhu et al. [7]	2024	Comparative Evaluation of YOLOv9, UNeXt and nnU-Net for Multiple Sclerosis Lesion Segmentation	Compared YOLOv9 variants, UNeXt, and nnU-Net on the MICCAI 2016 dataset.	UNeXt and nnU-Net outperformed YOLOv9, with reported Dice scores of approximately 0.79, 0.76, and 0.57, respectively.
Pishvai et al. [8]	2024	Advantages of Brain Parcellation in Multiple Sclerosis Lesion Segmentation	Diffusion-based segmentation framework combined with brain parcellation to incorporate anatomically meaningful regions.	Reported a Dice score of approximately 74.14% and showed improved robustness and sensitivity to small abnormalities.
Wiltgen et al. [9]	2024	LST-AI: A Deep Learning Ensemble for Accurate MS Lesion Segmentation	Ensemble of multiple 3D U-Net models using composite loss functions.	Achieved approximately 0.65 Dice on MSSEG-1 and showed better lesion detection, particularly for larger lesions.
Zhang et al. [10]	2024	Towards an Accurate and Generalizable	U-Net-based model with instance normalization	Improved generalization across scanners, datasets,

		Multiple Sclerosis Lesion Segmentation Model Using Self-Ensembled Lesion Fusion	and test-time self-ensembled lesion fusion.	contrast variations, and image artifacts; reported 68.2% Dice on ISBI 2015.
Rondinella et al. [11]	2023	Enhancing Multiple Sclerosis Lesion Segmentation in Multimodal MRI Scans with Diffusion Models	Diffusion-model-based segmentation framework using multimodal MRI.	Demonstrated the potential of diffusion models for multimodal MS lesion segmentation and improved handling of subtle lesion characteristics.

2.2.1 Similar Applications

Several software tools and deep learning frameworks which have been developed to support automated Multiple Sclerosis (MS) lesion segmentation from magnetic resonance imaging (MRI). These applications provide practical and methodological references for the proposed segmentation framework, particularly in multimodal MRI analysis, three-dimensional deep learning, feature fusion, and robust lesion detection.

Rondinella et al. [15] suggested an attention based U-Net network with convolutional long short-term memory and attention modules for the segmentation of MS lesions. This method demonstrated the potential of recurrent contextual modeling for lesion segmentation and achieved an average Dice score of about 84% on the ISBI 2015 data. ConvLSTM was used to capture dependencies between spatial representations, while the attention components emphasized relevant lesion features. This paper thus validates the need for contextual information in addition to the local information learned from the traditional convolutions.

Hashemi et al. [16] studied the segmentation of MS lesions using FLAIR and T2-weighted MRI with modified U-Net and Attention U-Net architectures. The information from the different modalities was combined to create the final segmentation, and the study reported a Dice score of 81%. Results suggest complementary MRI contrasts can be helpful for lesion characterization especially when a lesion is hard to differentiate in a particular sequence. This multimodal approach is closely related to the motivation for using T1- and T2-weighted MRI as auxiliary modalities in the present research.

Shoeibi et al. [17] gave a detailed review on automated MS analysis using magnetic resonance imaging (MRI) by deep learning techniques. The review highlighted the transition from data-driven towards deep learning approaches and the potential of artificial intelligence for automated analysis of MRI data relevant to MS. It also identified a number of challenges, such as scarcity of data sets, MRI acquisition variability and the difficulty of obtaining reliable generalization from one condition to another. This work will be helpful in understanding the increasing use of deep learning in MS image analysis

and the need for reliable automated approaches.

To segment MS lesions, Zhang et al. [18] proposed ALL-Net, an anatomical-information and lesion-wise-loss integrated neural network. They used distance transformation mapping to include anatomical coordinate information and a lesion-wise loss function to mitigate the imbalance between small and large lesions. The method was able to obtain a high challenge score on the ISBI 2015 challenge and demonstrated improvement over a few publicly available segmentation tools on a larger Cornell MS dataset. A key element of this work is that it is explicitly targeted towards small-lesion detection, thus demonstrating that lesion-level considerations can be made that may not be captured by conventional voxel-wise optimization alone.

Aslani et al. [19] suggested multi-branch CNN model for segmentation of MS lesion. The architecture was designed to handle complementary information using separate branches, and to create the lesion segmentation with multi-scale feature fusion. Instead of using only one processing pathway, the researchers investigated whether multiple feature branches could extract complementary information from the multimodal MRI data. Such a strategy could be applied to the future in multimodal segmentation research as an illustration of the potential benefit of the combination of different MRI sequence information and feature representations.

Table 2.2: Similar Application

Application	Type	Main Input	Main Approach	Why relevant
Rondinella et al. [15]	Recurrent & Contextual Segmentation	Multi-modal MRI (FLAIR, T2, T1)	Attention-based U-Net combined with ConvLSTM	Captures spatial continuity and contextual relationships across MRI slices, showing how recurrent modeling
Hashemi et al. [16]	Multimodal Fusion & Feature Combination	Dual Modality MRI (FLAIR + T2-weighted)	Modified U-Net & Attention U-Net architectures	Supports complementary MRI modality use
Shoeibi et al. [17]	Comprehensive Literature Review	Multi-modal MRI (FLAIR, T2, T1)	Attention-based U-Net combined with ConvLSTM	Captures spatial continuity and contextual relationships across MRI slices, showing how recurrent modeling

Zhang et al. [18]	Small-Lesion Detection & Loss Optimization	FLAIR	Recurrent residual blocks + dual attention gates	Relevant to residual learning, attention, and difficult lesions
Aslani et al. [19]	Generalizable deep-learning framework	Multimodal MRI	Instance normalization + test-time self-ensemble	Relevant to multimodal data, missing modalities, and domain variation

2.3 Gap Analysis

While significant advances have been made in automated Multiple Sclerosis (MS) lesion segmentation, there are still several drawbacks in the current methods that impact the accuracy, robustness and applicability of automated methods. Although T1 and T2 weighted MRI images may offer complementary anatomical and lesion information, many of the recent methods focus on a single modality, specifically FLAIR. Multimodal methods exist but some methods fuse modalities via simple concatenation or prediction-level fusion, where the amount of input from each modality is not well controlled. Further, small and low-contrast lesions are still hard to segment, and the performance may be widely variable between the different lesion sizes, MRI acquisition conditions, scanners and patient populations. Recent research also shows that domain shift and missing MRI contrasts are still important issues to be addressed for real-world deployment.

Table 2.3: Identified Research Gaps and Proposed Methodological Responses

Aspect	Gap Finding	Proposed methodology
MRI Modality Utilization	Although T1-weighted and T2-weighted MRI provide complementary information, many MS lesion segmentation approaches have focused mainly on FLAIR. FLAIR is very useful for the visualization of lesions and T1 and T2 help to resolve in areas of ambiguity.	Use FLAIR as the primary modality while incorporating T1- and T2-weighted MRI as auxiliary modalities to provide complementary information.
Multimodal Fusion Strategy	Multimodal methods are already in place, such as: direct concatenation of features, output fusion or independent processing of modalities. Not all such approaches are able to selectively control the contribution of each modality as a function of the lesion context.	Introduce gated feature fusion so that the network can adaptively control the contribution of auxiliary T1/T2 features while preserving the primary FLAIR representation.
Preservation of Strong FLAIR Representation	Since FLAIR is generally the most important sequence for identifying MS lesions, unrestricted multimodal fusion	Maintain a FLAIR-centered primary branch and apply bounded residual

	may risk weakening the primary lesion-sensitive representation.	correction, allowing T1/T2 information to refine rather than replace the FLAIR-based prediction.
Small and Low-Contrast Lesions	Multiple studies report that segmentation performance decreases for smaller lesions. Small, sparse, irregular, and low-contrast lesions remain difficult to distinguish from surrounding tissues.	Perform lesion-level evaluation in addition to global metrics and analyze performance across lesion-size groups to determine whether the proposed model improves difficult lesion detection.
Intensity and MRI Variability	Intensity variation due to scanner characteristics, acquisition protocols, and imaging conditions can be seen in MRI scans, which decreases the generalization of MRI scans across datasets and clinical sites. Recently, scanner-agnostic and domain-generalized approaches have been specifically studied.	Apply a standardized preprocessing pipeline consisting of N4 bias-field correction and non-zero voxel Z-score normalization before model inference.
Evaluation Beyond a Single Global Metric	Dice is a single number that does not completely describe lesion detection behaviour, especially when lesion size and lesion intensity vary. Some studies demonstrate good overall performance with decreased sensitivity for smaller lesions.	Evaluate Dice, Precision, Recall and IoU, together with lesion-level recall/detection analysis and subgroup analysis based on lesion size and FLAIR brightness.

2.4 Summary

Overall, significant advances have been achieved in automated Multiple Sclerosis (MS) lesion segmentation using CNN based, attention enhanced, Transformer based, multimodal, self-ensemble, diffusion, and federated learning. There are still significant challenges to be addressed. Small lesions and low contrast remain a challenge for accurate segmentation and the use of complementary MRI modalities is still an open question. Performance may also vary depending on the scanner, acquisition protocol and patient population. Recent benchmark results also demonstrate that the performance is quite varied depending on the architecture and evaluation conditions, and so it is necessary to carry out careful experimental design and comprehensive evaluation.

The studies examined in this review also suggest that FLAIR is still a valuable tool for the identification of MS lesions, and that both T1- and T2-weighted MRI can be used to complement information in cases of uncertainty. But, when multiple modalities are combined, it is not necessarily controlled how much influence each modality has on the final segmentation result. Meanwhile, lesion-level analysis reveals that small and difficult lesions can be challenging despite a relatively high overall segmentation score of the model.

Based on these observations, a gated residual multimodal framework which preserves FLAIR is developed in this research. FLAIR is kept as the main modality and T1/T2 MRI is used as auxiliary data with gated feature fusion and bounded residual correction. The baseline architecture is a FLAIR-only UNETR model, and the proposed architecture is assessed using global segmentation metrics, lesion-level analysis and subgroup analysis, according to lesion size and FLAIR brightness. Chapter 3 presents the detailed methodology and system design, and the experimental results and comparison with the baseline are presented in Chapter 4.

Chapter 3

Research Methodology

This chapter outlines the research methodology for the automated system for Multiple Sclerosis (MS) lesion segmentation from three-dimensional brain Magnetic Resonance Imaging (MRI) using deep learning. It includes entire system design, methodology pipeline, MRI data preprocessing, baseline and proposed model architectures, training and evaluation processes and functional and non-functional requirements during the development.

3.1 Methodology

3.1.1 Overview

The main objective of this study is to construct an automated framework for three-dimensional brain MRI volume-based Multiple Sclerosis (MS) lesion segmentation which is reliable. The manual delineation of MS lesions is laborious and can be subjective. This presents the need for an automated deep learning-based system that is consistently able to detect and segment the lesions and utilise information from complementary modalities, such as multimodal MRI.

The gated residual multimodal architecture of Experiment 7 Phase 2 is the main technical contribution of this research. In this context, FLAIR is employed as the main modality and additional information is obtained from T1 and T2 weighted MRI. The incorporation of complementary features from T1 and T2 is controlled by a gated fusion mechanism and keeps the representation acquired from FLAIR, which is lesion-sensitive. The additional multimodal information is then used in a bounded residual correction mechanism to adjust the initial segmentation performed by FLAIR.

A FLAIR-only UNETR model is first created as a baseline to evaluate the proposed architecture. This is used as a reference to compare the effectiveness of the proposed multimodal approach to a single-modality approach using Transformer. Both configurations are assessed in the same experimental settings with respect to quantitative segmentation metrics and analyses at the level of lesions.

A standardized MRI preprocessing pipeline is also provided in the methodology. First, the MR volumes are corrected for intensity non-uniformity using N4 bias-field correction, and then they are normalized to have non-zero Z-scores. In this step, only the non-zero voxels are used to calculate the mean and standard deviation, and the background (or zero) voxels are not modified. Using the experimental protocol at the patient level and three-dimensional MRI data, the models are then trained and evaluated. Measurement of performance is done by Dice Coefficient, Precision, Recall and Intersection over Union (IoU) with further analysis of lesion detection based on lesion size and FLAIR brightness.

3.1.2 Proposed Methodology/ System Design

The proposed methodology is implemented as an end-to-end three-dimensional MRI segmentation pipeline for automated Multiple Sclerosis (MS) lesion segmentation. The workflow begins with dataset organization and patient-level separation, followed by MRI preprocessing. The preprocessed data are then used to independently develop and train the FLAIR-only UNETR baseline and the proposed Gated Residual Multimodal Network. Both models use a common three-dimensional inference procedure, followed by comprehensive quantitative and lesion-level performance evaluation.

i. Dataset Acquisition and Preparation:

The study uses the publicly available MSLesSeg dataset, which contains three-dimensional brain MRI scans and corresponding lesion segmentation masks. Each timepoint provides three MRI modalities: FLAIR, T1-weighted, and T2-weighted images. The dataset is organized at both the patient and timepoint levels so that the MRI modalities and their corresponding ground-truth masks remain correctly associated. Patient-level separation is maintained between the training, validation, and testing data to reduce the risk of data leakage and provide a more reliable assessment of model generalization. The dataset contains 53 training patients and 22 test patients, with the test patients kept separate from model development.

ii. MRI Preprocessing Pipeline:

A two-stage intensity preprocessing pipeline is applied to the MRI volumes before model training. First, N4 bias-field correction is performed to reduce low-frequency intensity non-uniformity in the MRI scans. A foreground mask derived from non-zero voxels is used during this process. The corrected images are then subjected to non-zero voxel Z-score normalization, where the mean and standard deviation are calculated only from non-zero voxels. The background remains as zero during the normalization process. This yields standardized MRI inputs without affecting the spatial structure of the original MRI volumes. The dataset is already registered to a standard spatial representation of $182 \times 218 \times 182$ which is kept consistent throughout the subsequent processing.

iii. Baseline Model Development:

The baseline architecture model is developed first, which is a FLAIR-only UNETR model. FLAIR is used as the sole input modality because it provides strong contrast for MS lesions and serves as the primary sequence for lesion identification. The baseline establishes a reference against which the contribution of T1- and T2-weighted information and the proposed multimodal fusion strategy can be evaluated. UNETR is a Transformer-based three-dimensional segmentation architecture that combines a Transformer encoder with a CNN-based decoder through skip connections.

iv. Proposed Gated Residual Multimodal Architecture:

The proposed architecture extends the baseline by incorporating complementary information from multiple MRI modalities while keeping FLAIR as the primary input. It consists of a primary FLAIR branch and an auxiliary T1/T2 branch. The FLAIR branch generates the main lesion representation, while the T1- and T2-derived features provide additional information from the other MRI sequences. A gated feature-fusion mechanism controls the contribution of the auxiliary features before they are combined with the

primary representation. A bounded residual correction mechanism then refines the FLAIR-based segmentation using the additional multimodal information. In this way, the auxiliary modalities are used to refine the primary prediction rather than replace the representation obtained from FLAIR.

v. Model Training and Optimization:

The baseline and proposed models are trained using the preprocessed three-dimensional MRI volumes and their corresponding ground-truth lesion masks. Training samples are generated using the established three-dimensional sampling strategy, and augmentation is applied to the training data to improve robustness and reduce overfitting. Each model is optimized using the loss configuration defined for its respective experiment. The checkpoint with the strongest validation performance is retained for subsequent testing.

vi. Three-Dimensional Inference and Segmentation:

After training, the selected checkpoint is used to generate lesion-probability volumes from unseen MRI data. Because three-dimensional MRI volumes are fairly large, sliding window inference is performed to process the images in smaller chunks and then the prediction for the whole volume is built. In the new multimodal setup, the main input is still FLAIR with T1 and T2 inputs being complementary. Then, the created probability map is transformed into a binary lesion mask by the chosen segmentation threshold.

vii. Performance Evaluation and Lesion-Level Analysis:

The evaluation of the segmentation results is carried out by comparing the predicted masks with the available ground-truth masks. Dice Coefficient, Precision, Recall and Intersection over Union (IoU) are the primary evaluation metrics used. In addition to global voxel-level evaluation, lesion-level analysis is carried out to explore the ability of the models to identify individual lesions. Model behavior is explored in small lesions, low-contrast lesions and other difficult lesion groups by considering the lesion size and FLAIR brightness. Several evaluation measures are used since there is no single overlap measure that is sufficient to describe the performance of segmentation. The MSLesSeg dataset also focuses on multiple evaluation measures to highlight various aspects of segmentation performance, such as overlap, sensitivity, precision, and lesion-level behavior.

3.1.3 Functional and Nonfunctional Requirements

It is crucial to have a clear definition of functional and non-functional requirements to guide the development and evaluation of the proposed Multiple Sclerosis (MS) lesion segmentation framework. The functional requirements specify the main tasks that the system is expected to perform, including MRI data input, preprocessing, model inference, lesion segmentation, result visualization, and quantitative performance evaluation. The non-functional requirements describe the desired characteristics of the framework, including performance, reliability, reproducibility, computational efficiency, scalability, and maintainability.

Functional Requirements:

The functional requirements of the proposed MS lesion segmentation framework are summarized in Table 3.13.

Table 3.1.3: Functional Requirements

No	Requirement	Description
1	Multimodal MRI Input Processing	The system must support three-dimensional brain MRI volumes in NIfTI format. The FLAIR-only UNETR baseline uses FLAIR as its input, whereas the proposed Gated Residual Multimodal Network uses FLAIR as the primary modality together with T1- and T2-weighted MRI as auxiliary modalities.
2	MRI Preprocessing	The system must preprocess MRI volumes using N4 biasfield correction followed by non-zero voxel Z-score normalization. The mean and standard deviation are calculated only from non-zero foreground voxels, while the zero-valued background is preserved.
3	Model Loading and Selection	The system must be capable of loading the trained FLAIR-only UNETR baseline and the trained Experiment 7 Phase 2 Gated Residual Multimodal Network checkpoints for segmentation and comparative evaluation.
4	3D MS Lesion Segmentation	The system must generate a three-dimensional MS lesion probability map from the supplied MRI volume and convert the resulting prediction into a binary lesion segmentation mask using the selected segmentation threshold.
5	3D Sliding-Window Inference	The system must support three-dimensional slidingwindow inference so that complete MRI volumes can be processed through manageable regions of interest and the patch-wise predictions can subsequently be reconstructed into a complete three-dimensional prediction volume.
6	Segmentation Visualization	The system must provide visualization of the FLAIR MRI, ground-truth lesion mask when available, and predicted lesion segmentation. Ground-truth and predicted lesion regions should be displayed as overlays on the corresponding FLAIR MRI for visual inspection.
7	Performance Evaluation	The evaluation pipeline must calculate the principal segmentation metrics, including Dice Coefficient, Precision, Recall, and Intersection over Union (IoU), by comparing predicted lesion masks against the corresponding ground-truth masks.
8	Lesion-Level Analysis	In addition to global voxel-level evaluation, the system must support lesion-level analysis, including individual lesion detection and lesion-level recall. It must also allow segmentation performance to be investigated according to

		lesion size and FLAIR brightness to identify challenging lesion groups.
9	Baseline and Proposed Model Comparison	The framework must support a direct comparison between the FLAIR-only UNETR baseline and the proposed Gated Residual Multimodal Network under the established evaluation procedure to determine the effect of multimodal feature integration.

Nonfunctional Requirements:

The nonfunctional requirements define the operational and quality characteristics expected from the proposed framework. They are summarized in Table 3.2.

Table 3.1.3: Nonfunctional Requirements:

No	Requirements	Description
1	Performance	The framework should efficiently process three dimensional MRI volumes using GPU acceleration when available. Sliding-window inference should allow full-resolution MRI volumes to be segmented without requiring the entire volume to be processed by the network simultaneously.
2	Reproducibility	Dataset separation, training configuration, model parameters, preprocessing operations, and experimental settings should remain consistent across experiments so that the baseline and proposed models can be compared under controlled conditions.
3	Memory Efficiency	The framework should process three-dimensional MRI data using manageable regions of interest and batch sizes appropriate for the available computational resources. Sliding-window inference and patch-based processing should reduce GPU memory requirements associated with full-volume 3D segmentation.
4	Reliability	The system should validate required MRI inputs before inference and handle missing or invalid input files without terminating unexpectedly. For the multimodal model, the required FLAIR, T1, and T2 volumes should be correctly associated before segmentation is performed.

5	Data integrity	Patient-level separation must be maintained between model-development and testing data to minimize data leakage. MRI modalities and corresponding ground-truth masks belonging to the same patient and timepoint must remain correctly associated throughout preprocessing, training, and evaluation.
6	Scalability	The modular framework should allow additional MRI modalities, segmentation architectures, preprocessing techniques, or experimental configurations to be incorporated in future studies without requiring complete redesign of the pipeline.
7	Portability	The implementation should operate in a Pythonbased deep learning environment and support CUDA-enabled GPU execution when compatible hardware is available. The framework should retain the ability to execute essential operations on CPU where technically supported, although with increased computational time.
8	Maintainability	The dataset handling, preprocessing, model definition, training, inference, and evaluation parts should be modular, allowing for easy swapping out or modification of individual components without the need to rewrite the entire segmentation framework.

3.1.4 Data flow Diagram:

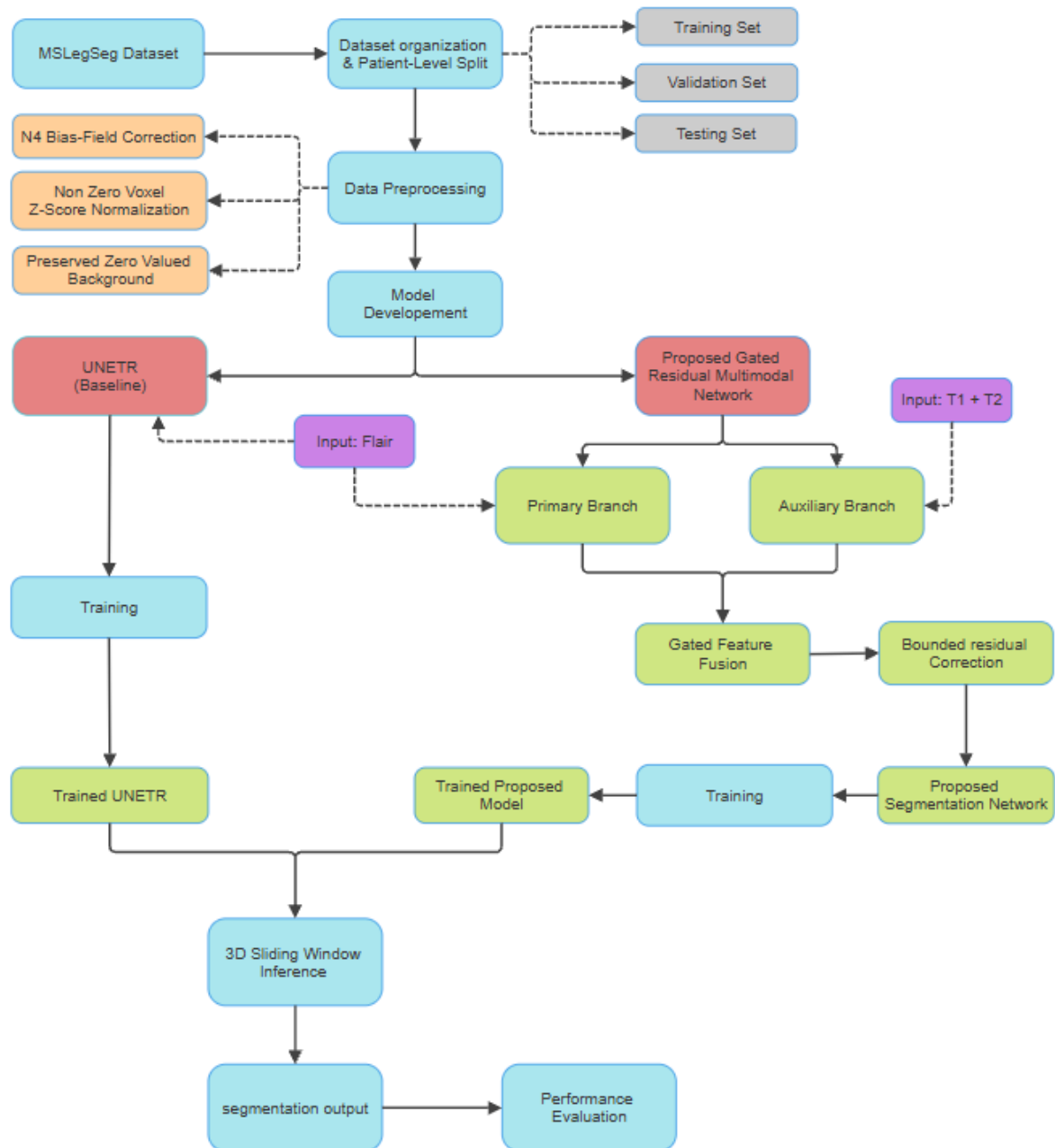


Figure 3.1.4 Data flow diagram

3.1.5 UI Design

The proposed system includes a web-based user interface that provides a simple way to perform automated Multiple Sclerosis (MS) lesion segmentation from three-dimensional brain MRI volumes. Although the main focus of the project is the development and evaluation of the segmentation models, the interface provides a practical environment for loading MRI data, running segmentation, and visually examining the predictions.

The interface allows users to upload the required MRI volumes in NIfTI format, including FLAIR, T1-weighted, and T2-weighted images for the proposed multimodal model. A corresponding ground-truth lesion mask can also be provided when available for evaluation and visualization. The system supports both the FLAIR-only UNETR baseline and the proposed Gated Residual Multimodal Network, allowing users to examine their segmentation outputs according to the required input configuration.

The primary sequence displayed for visualization is FLAIR MRI and the ground truth and predicted lesions can be overlaid and compared. The user can switch between each of the axial slices of the 3D volume and can display the segmentation outcome at various anatomical locations. The interface also has controls for setting the segmentation threshold and resetting the uploaded data if necessary.

The interface is structured and consistent, with the Frontend created with Gradio and custom CSS & JavaScript. It works with the Python deep learning pipeline consisting of the preprocessing, model loading, three dimensional sliding window inference and segmentation with the configured backend components.

The interface is designed to primarily be used for research and experimental assessment and not used for stand-alone clinical diagnosis. It enables an easy presentation of the segmentation framework, comparison of predictions made by the baseline and proposed model, and visual inspection of the segmentation results (lesion masks). Batch processing of MRIs, additional visualization features, automated report generation, and the integration of the component into larger clinical research processes are additional features that will be added in the futures. All the UI feature added in figure 3.1.4.

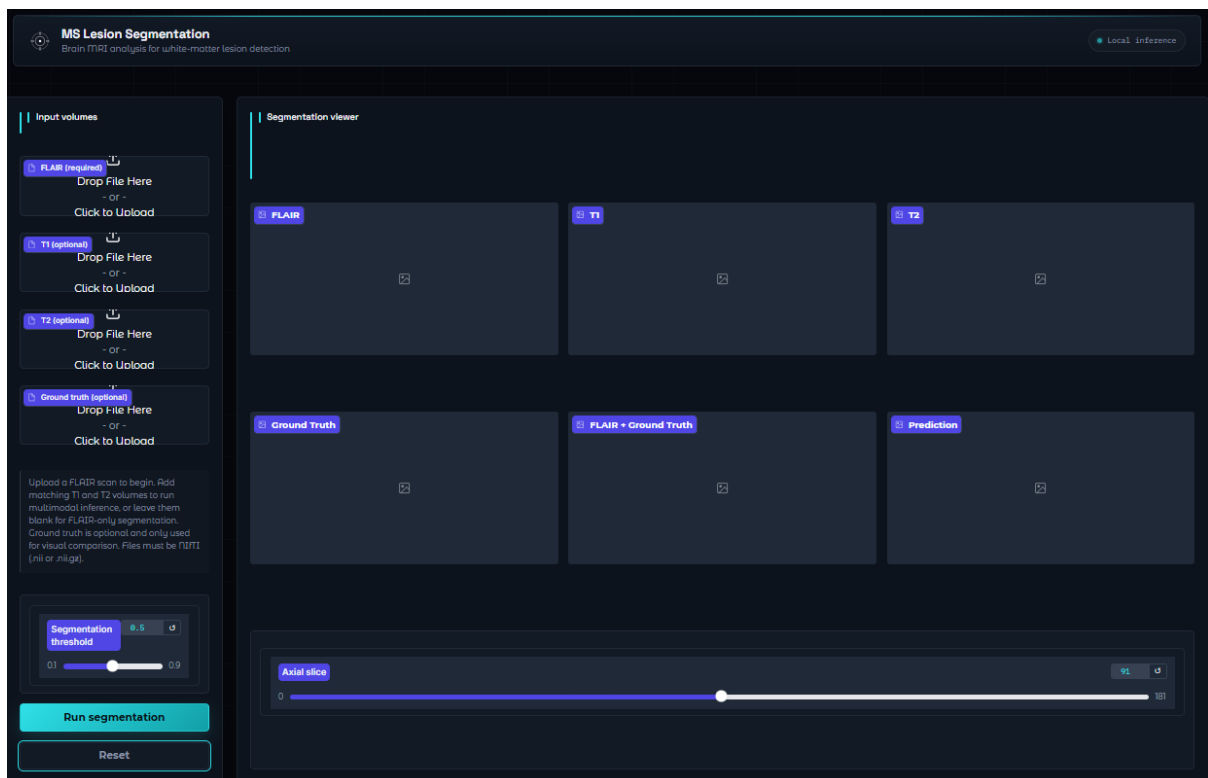


Figure 3.1.4: Interface home page

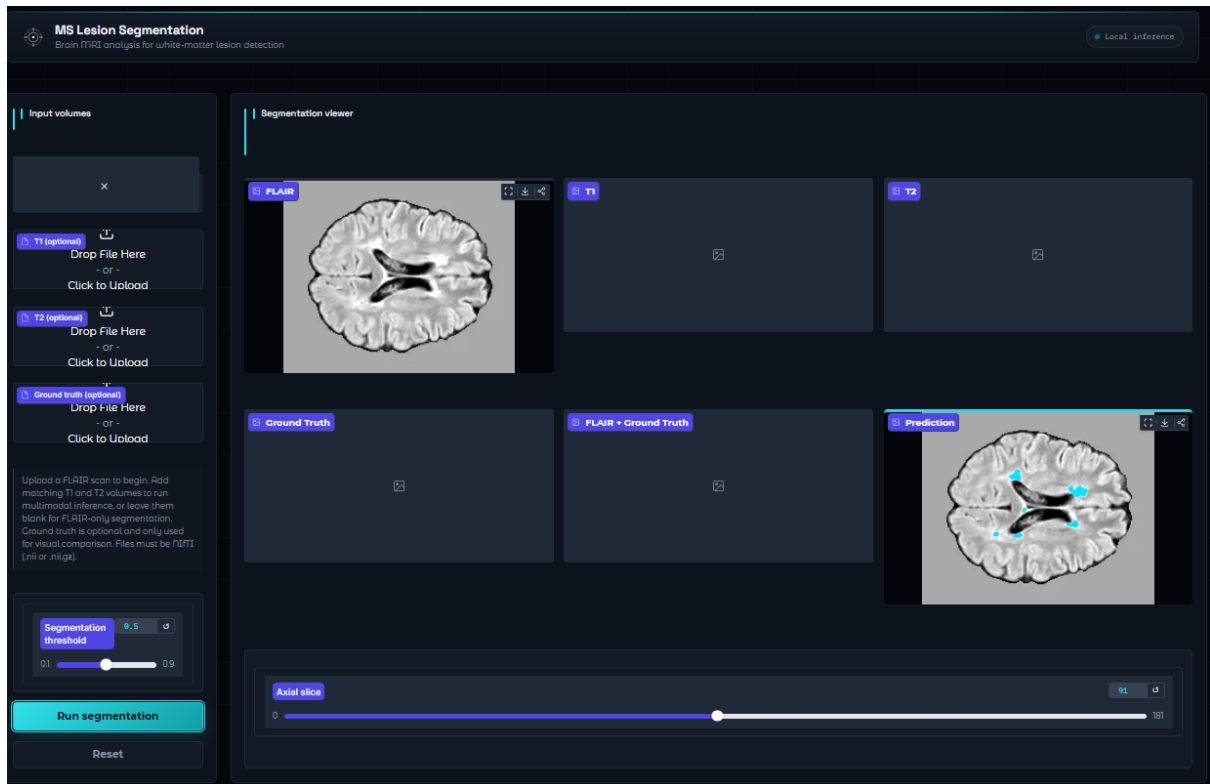


Figure 3.1.4 : Only Flair Input

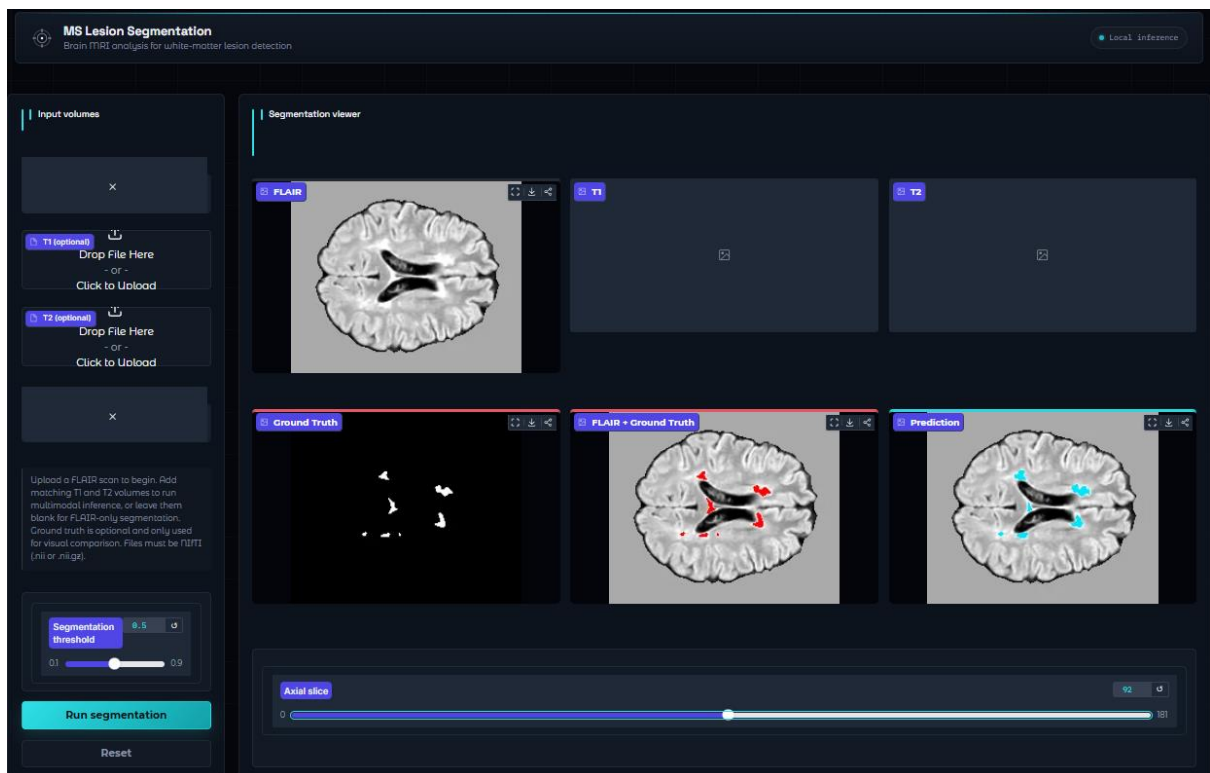


Figure 3.1.4 : Flair With Ground Truth

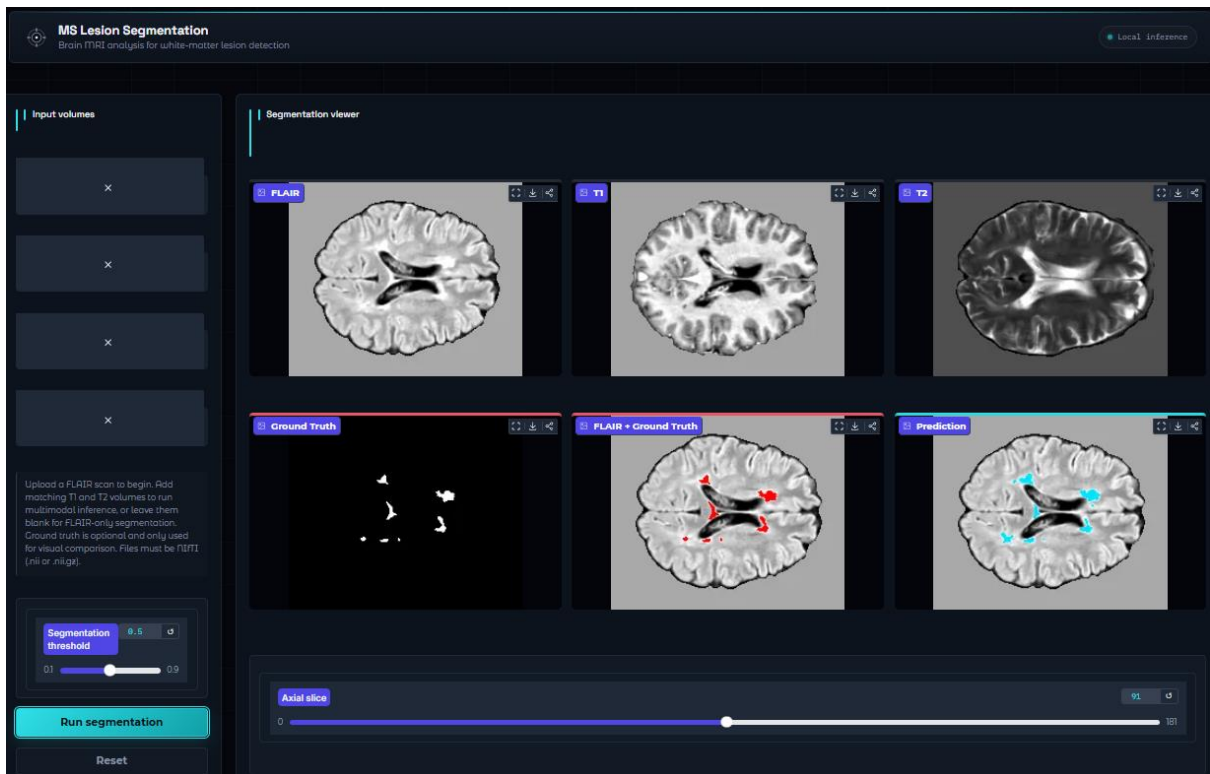


Figure 3.1.4: Flair T1 and t2 Input and Segmentation

3.2 Detailed Methodology and Design

3.2.1. Data Collection and Organization

The three dimensional longitudinal MRI dataset MSLesSeg is publicly available and is used in the study for Multiple Sclerosis (MS) lesion segmentation. It offers three complementary MRI modalities (FLAIR, T1-weighted and T2-weighted images) and corresponding lesion segmentation masks. The MRI volumes are supplied in NIfTI format and are structured by patient and time point, each matched with the lesion mask for the modality. A total of 75 patients were included in the dataset, 53 in the training/development set and 22 as an independent test set. The experimental pipeline preserves patient separation as some patients have multiple MRI timepoints. This eliminates the possibility of having scans from the same patient in separate subsets and minimises the risk of leakage of information during model development and evaluation.

Table 3.2.1: MSLesSeg Dataset Overview

Category	Information
Dataset	MSLesSeg
Total Patients	75
Training Patients	53
Independent Test Patients	22
MRI Modalities	FLAIR, T1-weighted, T2-weighted
Ground Truth	MS lesion segmentation mask
Data Type	3D NIfTI MRI volumes
Standard Volume Representation	182 x 182 x 182

Data Organization	Patient → Timepoint → MRI modalities + mask
Primary Modality	Flair
Auxiliary Modalities	1-weighted and T2-weighted

Further patient-level splitting of the 53 development patients is done for training and validation. All 22 test patients are entirely separated from the model training and the model selection and are only used for the final evaluation. Multiple timepoints from the same patient will not be in different subsets, which helps reduce the risk of patient-level data leakage and allows for a more accurate evaluation of model performance.

The organization of the set of data is as follows:

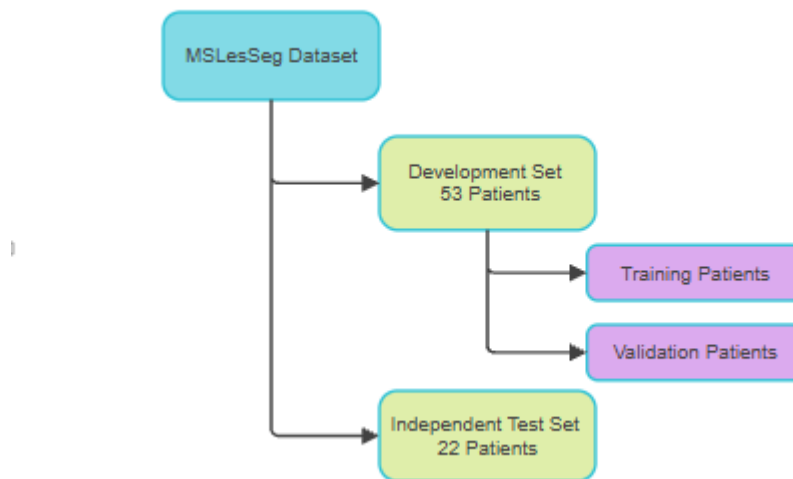


Figure 3.2.1: dataset organization

Each patient and time point has an associated MRI modalities and lesion mask that is stored as a unit of data:

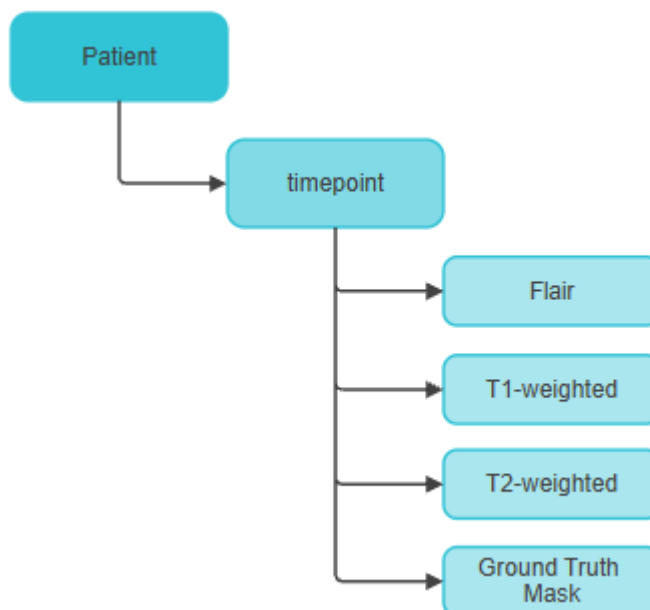


Figure 3.2.1: Timepoints Structure

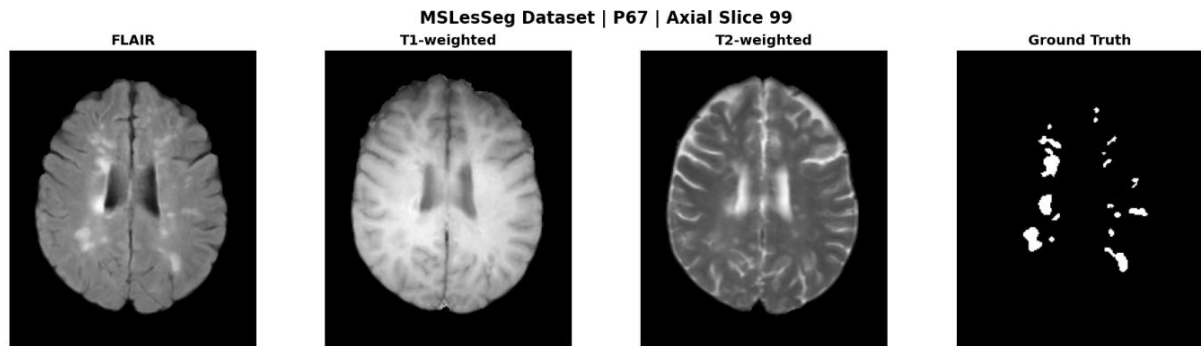


Figure 3.2.1: Data Slice Structure

The two experimental configurations assessed in this study are directly supported by this organization. The FLAIR-only UNETR baseline takes the FLAIR volume as its input, but the proposed Gated Residual Multimodal Network takes FLAIR as the primary modality and T1- and T2-weighted images as auxiliary inputs. Supervised training is done on both configurations with the associated ground-truth lesion masks.

3.2.2. Preprocessing

The MRI preprocessing step aims at enhancing the uniformity of the input images but retaining the anatomical and spatial data that would be required to segment Multiple Sclerosis (MS) lesions. The MRI intensity can be inconsistent due to intensity inhomogeneity due to acquisition and also variations in intensity scales among volumes. To cope with these variations, two consecutive preprocessing steps are used: N4 bias-field correction and non-zero voxel Z-score normalization. Each MRI volume is preprocessed independently and applied to FLAIR, T1-weighted and T2-weighted modalities. The ground-truth lesion masks are not intensity-normalized or otherwise processed and are stored in their binary form.

i. N4 Bias-Field Correction

The first pre-processing step is N4 bias field correction, which involves independently correcting each MRI modality for global smooth low frequency intensity inhomogeneity within MRI volume. Such variations can result in different intensity levels in the same areas of tissue in different images, and can make the segmentation model more sensitive to intensity differences between images that occur due to scanner or acquisition differences.

Because the data set has background regions in which the value is zero, a mask is created for each MRI volume using the non-zero voxels. This mask is then used in the N4 correction to correct only the relevant brain region and not the background. The original data are not modified; the corrected volumes are stored in the N4_preprocessed directory. Histograms of intensity distribution of an original FLAIR volume and its N4-corrected version are compared to investigate the effect of the correction. This comparison

illustrates the re-distribution of the intensities in the foreground after bias field correction.

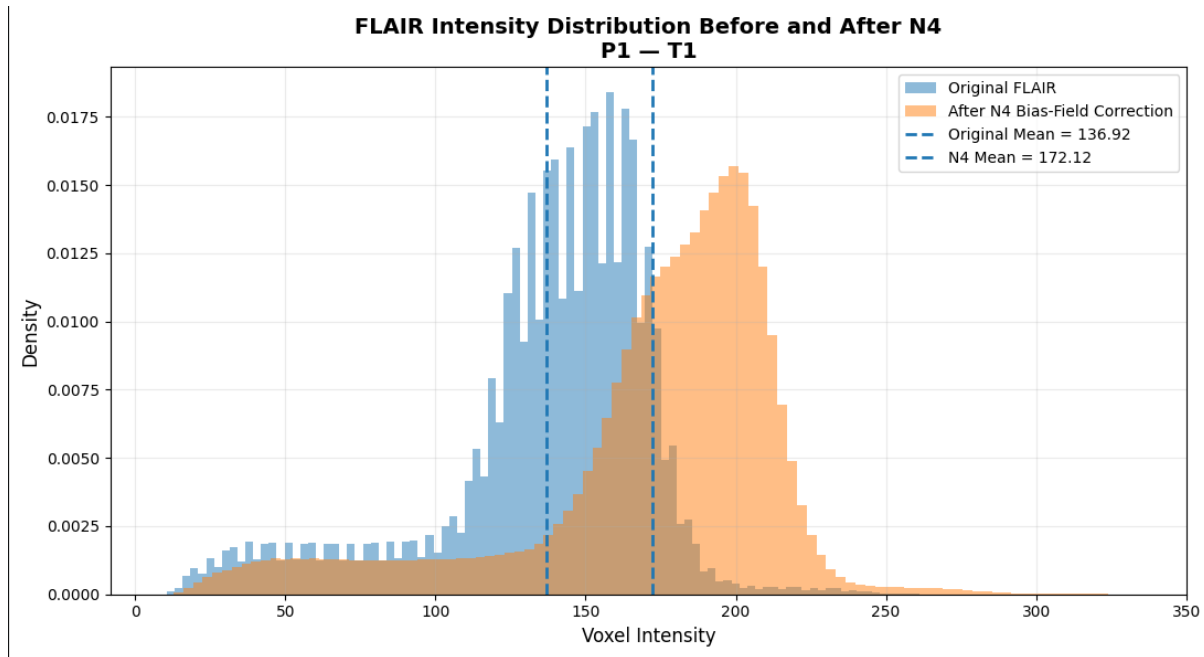


Figure 3.2.2: FLAIR intensity distribution before and after N4 bias-field correction for P1–T1.

After N4 correction the intensity distribution changes as can be seen in the histogram. The mean foreground intensity of the original image for representative P1-T1 FLAIR volume was about 136.92 and for the N4 corrected image was about 172.12. This change is due to the change in intensity field caused by the correction process and not due to changes in the anatomical structure of the MRI volume.

ii. Non-Zero Voxel Z-Score Normalization

After N4 correction, Z-score normalization was done to normalize the intensities in each MRI volume. Only non-zero voxels were used for the normalization, and the background voxels that have a value of zero were expressly kept as zero.

The mean intensity μ and standard deviation σ of the non-zero voxels were computed for each volume. The non-zero voxels xx were then transformed as follows:

$$Z = \frac{x - \mu}{\sigma}$$

Where z represents the standardized voxel intensity.

Using only non-zero voxels prevents the large number of background voxels from affecting the calculated mean and standard deviation. As a result, the normalization reflects the intensity distribution of the brain region rather than a combination of brain and background values. The processed images were stored in the N4_Zscore_preprocessed directory.

The expected result of this operation is that the foreground intensity distribution approaches a mean of approximately 0 and a standard deviation of approximately 1 for each MRI volume, while the zero-valued background remains unchanged.

iii. Visual Verification of the Preprocessing Pipeline

To verify that preprocessing did not alter the spatial or anatomical information of the MRI scans, the same axial slice from the original, N4-corrected, and N4 + Z-score normalized FLAIR volumes was visualized.

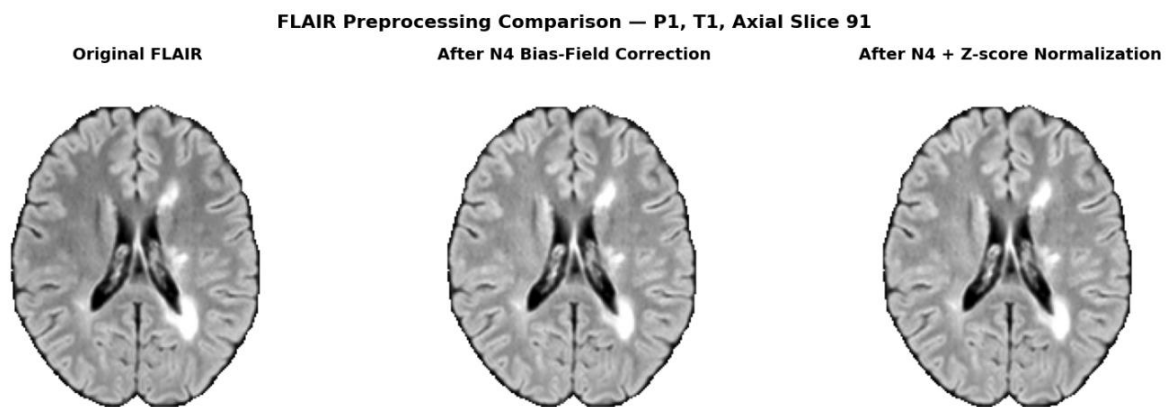


Figure 3.2.2: Visual comparison of the original, N4-corrected, and N4 + Z-score normalized FLAIR images for P1–T1, axial slice 91.

As shown in the figure, the anatomical structures and overall spatial arrangement of the brain remain consistent across all three preprocessing stages. The lesion-related hyperintense regions visible in the FLAIR image are also preserved in their original locations. The main differences between the stages are therefore related to voxel intensity characteristics rather than changes in spatial geometry.

This verification is important because the preprocessing pipeline is intended to improve intensity consistency while preserving the location and shape of the anatomical structures and MS lesions.

iv. Dataset-Level Verification of Z-Score Normalization

In addition to the representative MRI visualization, the effect of normalization was examined across the available patient timepoint volumes. The mean foreground intensity was calculated for each volume before preprocessing, after N4 correction, and after N4 + Z-score normalization. The mean foreground intensity of each volume was computed before, after N4 correction and after N4 + Z-score normalization.

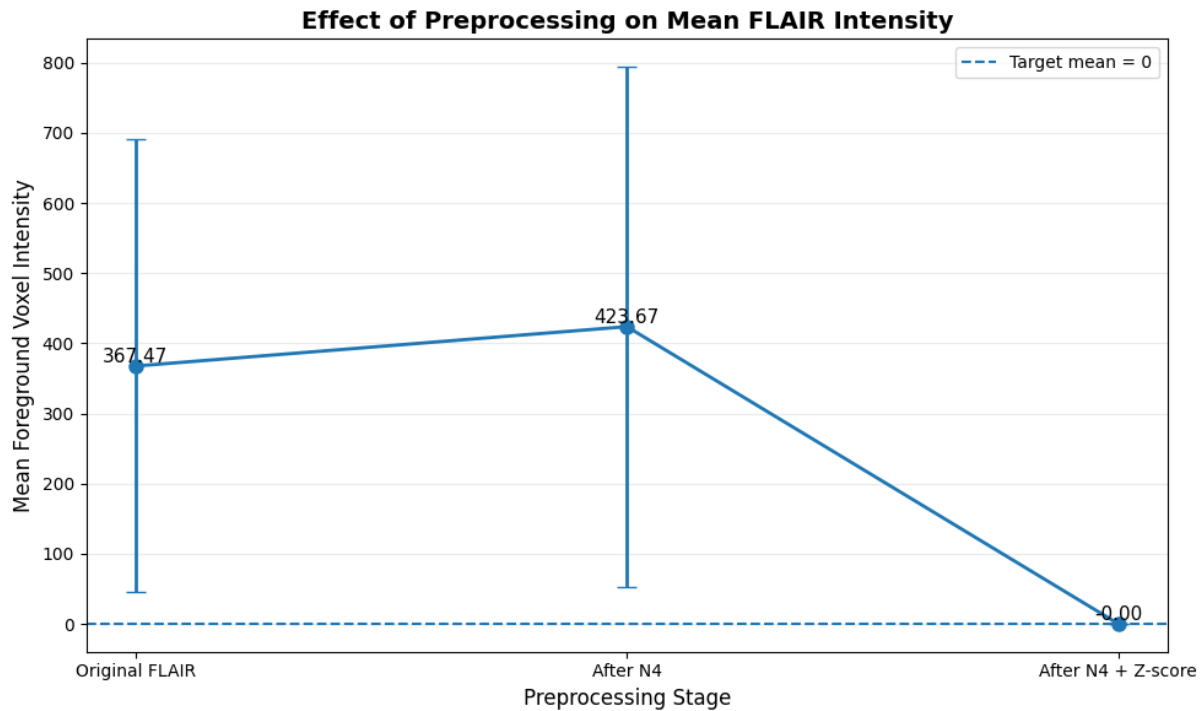


Figure 3.2.2: Effect of preprocessing on mean FLAIR intensity across the MRI volumes.

The graph shows how the mean foreground intensity changed from the three steps of the preprocessing. The original and N4-corrected volumes have their own intensity scales, while the mean intensity after N4 correction and z score normalization is centred near to 0. This validates the desired effect of standardization by the Z-score transformation.

The standard deviation values corresponding to them were also checked during the verification process. The standard deviation of the non-zero voxel intensities is close to 1 after Z-score normalization, in line with the mathematical goal of the normalization procedure.

3.2.3 Patient-Level Data Splitting

The dataset was split at the patient level instead of the image/slice level to guarantee a robust assessment of the proposed models for MS lesion segmentation. All MRI scans and timepoints within a patient were grouped together into a data partition, as the MSLesSeg data set contains multiple MRI timepoints for some of the patients. This will ensure that data from the same patient will not be present in both the training and test sets, and minimise the chances of data leakage.

The dataset was organized into two main groups: a training/development group containing patients P1–P53 and a held-out test group containing patients P54–P75. The training group contains multiple patient-timepoint scans, while the test group consists of the designated held-out patients. In total, the training portion contains 93 patient-timepoint scan pairs, whereas the test portion contains 22 patients.

For model development, the training/development data were further divided into training and validation subsets. The validation subset was used to monitor model performance and select the best-performing checkpoint, while the held-out test patients remained

completely separate until the final evaluation. Therefore, no scan or timepoint from patients P54–P75 was used during model training or validation.

Patient level splitting is especially relevant to longitudinal medical imaging where multiple scans from the same patient may exhibit similar anatomical features, lesion patterns, and acquisition related properties. Randomly dividing individual scans, slices, or patches could allow information from the same patient to appear in both training and evaluation data, potentially leading to an overly optimistic estimate of model performance. Maintaining patient separation throughout the experimental pipeline therefore provides a more reliable assessment of how well the models generalize to previously unseen patients.

3.2.4 Three-Dimensional Training Sample Generation

Since the proposed system performs three-dimensional MRI lesion segmentation, the volumetric MRI scans are processed as 3D training samples rather than being converted into individual 2D slices. Each sample contains the three MRI modalities—FLAIR, T1-weighted, and T2-weighted images—along with the corresponding binary ground-truth lesion mask.

The preprocessed MRI volumes are divided into smaller three-dimensional regions using a patch-based sampling strategy. Each training sample has a spatial size of $96 \times 96 \times 96$ voxels. Processing smaller regions instead of complete brain volumes reduces GPU memory requirements while retaining the three-dimensional spatial and contextual information surrounding MS lesions.

To increase the likelihood of selecting lesion-containing regions, random positive and negative patch sampling is used with a positive-to-negative ratio of 4:1. In each sampling operation, four 3D patches are generated. This strategy provides the models with more lesion-containing examples, which is useful because lesions occupy only a small portion of the complete brain volume.

The generated patches are converted into tensors and provided to the segmentation networks during training. For the FLAIR-only UNETR baseline, only the FLAIR volume is used as input. The proposed multimodal model receives FLAIR as the primary input, with T1- and T2-weighted information provided as auxiliary inputs. The corresponding ground-truth lesion mask serves as the target for supervised learning.

This patch-based 3D training strategy allows the models to learn volumetric spatial relationships between neighboring voxels and lesion structures while keeping computational and GPU memory requirements within manageable limits.

The flow is like this:

Full MRI Volume $\rightarrow$ 3D Patch Sampling $\rightarrow$ $96 \times 96 \times 96$ Patch $\rightarrow$ Model Input

All the procedure shows in Figure 3.2.4.

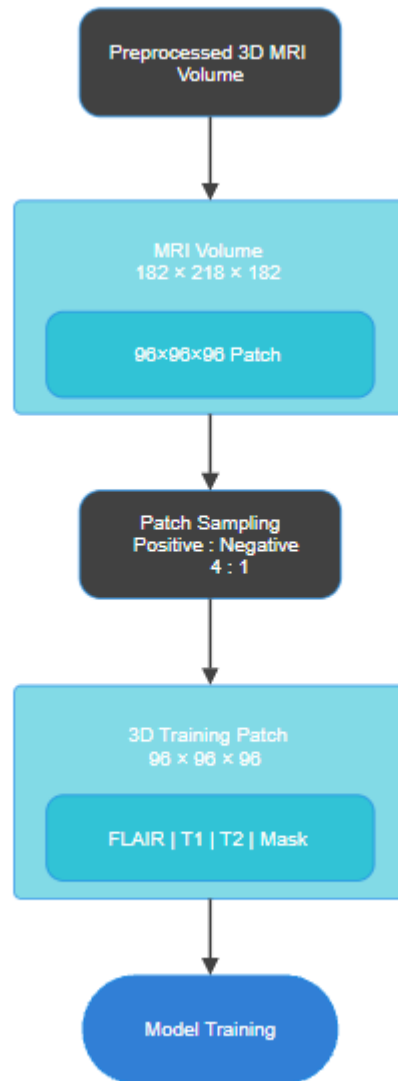


Figure 3.2.4: Three-dimensional patch generation from preprocessed MRI volumes.

3.2.5 FLAIR-Only UNETR Baseline

To serve as a reference performance for comparison, a FLAIR-only UNETR model was developed as the baseline model for MS lesion segmentation. FLAIR was chosen as the only input modality as MS lesions typically show hyperintensity on FLAIR images which is important for lesion identification. The baseline represents a controlled reference to be used to assess the added value of the other MRI modalities introduced in the proposed architecture.

The UNETR (UNet Transformers) architecture was employed as a three-dimensional segmentation network. Unlike conventional convolutional U-Net architectures, UNETR uses a Transformer-based encoder to capture long-range spatial relationships within volumetric MRI data. The encoder processes the 3D input and extracts contextual representations, while the decoder progressively restores the spatial resolution to produce the final lesion segmentation.

In this experiment, only the preprocessed FLAIR volume was provided as the network input, while the corresponding binary ground-truth mask was used as the target for

supervised training. The 3D training patches generated using the procedure described in Section 3.2.4 were used as model inputs. The baseline was therefore intentionally kept as a single-modality configuration, allowing its performance to be compared with the proposed model using FLAIR, T1-weighted, and T2-weighted MRI.

During training, the UNETR model learned to predict a binary MS lesion mask from the FLAIR input. After training, the selected model was used for whole-volume prediction through three-dimensional sliding-window inference. The complete MRI volume was processed as a series of overlapping 3D regions, and the individual predictions were combined to reconstruct the final lesion segmentation for each test patient. The FLAIR-only UNETR serves as the baseline experiment and provides a controlled reference for determining whether incorporating complementary T1- and T2-weighted information through the proposed gated residual multimodal architecture can improve lesion segmentation, particularly for difficult lesion groups.

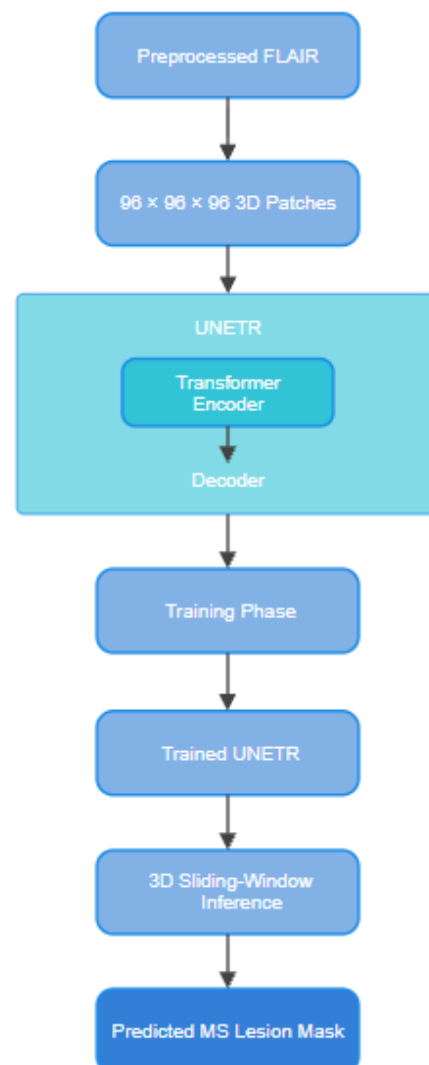


Figure 3.2.5: Workflow of the FLAIR-only UNETR baseline for MS lesion segmentation.

3.2.6 Proposed Gated Residual Multimodal Architecture

To incorporate complementary information from multiple MRI modalities, a Gated Residual Multimodal Network was developed as the proposed segmentation architecture. The model uses FLAIR as the primary modality, while T1- and T2-weighted MRI are used as auxiliary modalities. This design is based on the complementary roles of the MRI sequences: FLAIR provides strong lesion-related intensity information, while T1- and T2-weighted images provide additional anatomical and tissue-related information.

The proposed architecture consists of a primary FLAIR branch and an auxiliary T1/T2 branch. The FLAIR image is processed through the primary branch to extract lesion-related features. In parallel, the T1- and T2-weighted images are processed through the auxiliary branch to obtain complementary feature representations. Keeping the two pathways separate allows the model to retain FLAIR as the main source of lesion information while incorporating additional information from the other modalities.

The features extracted from the two branches are then combined using a Gated Feature Fusion mechanism. Rather than directly concatenating the modality features, the gating mechanism determines how much information from the auxiliary branch should be incorporated into the FLAIR representation. This is a controlled means of leveraging the complementary multimodal information and helps decrease the chances that any irrelevant or conflicting features will dominate the segmentation process.

Feature fusion is followed by a Bounded Residual Correction machine, which corrects the fused representation. The residual part adds another correction to the main feature representation but limits the size of this correction. This enables the information obtained from the auxiliary modalities to be used to modify the FLAIR-based representation, but not entirely replace the features extracted from the FLAIR.

The refined representation is then passed through the segmentation network to generate the final binary MS lesion prediction. The overall processing flow can be summarized as:

FLAIR primary input → Primary feature extraction
T1 + T2 auxiliary inputs → Auxiliary feature extraction
Primary + auxiliary features → Gated Feature Fusion → Bounded Residual Correction
→ Segmentation Network → Lesion Mask

The proposed architecture is trained using the same patient-level data partition and 3D training-sample generation strategy described in the preceding sections. Using the same experimental setup provides a consistent basis for comparing the multimodal model with the FLAIR-only UNETR baseline.

The main purpose of the proposed architecture is to examine whether selectively incorporating complementary T1- and T2-weighted information can improve the lesion representation obtained from FLAIR alone, particularly for lesions that are difficult to characterize based only on FLAIR intensity.

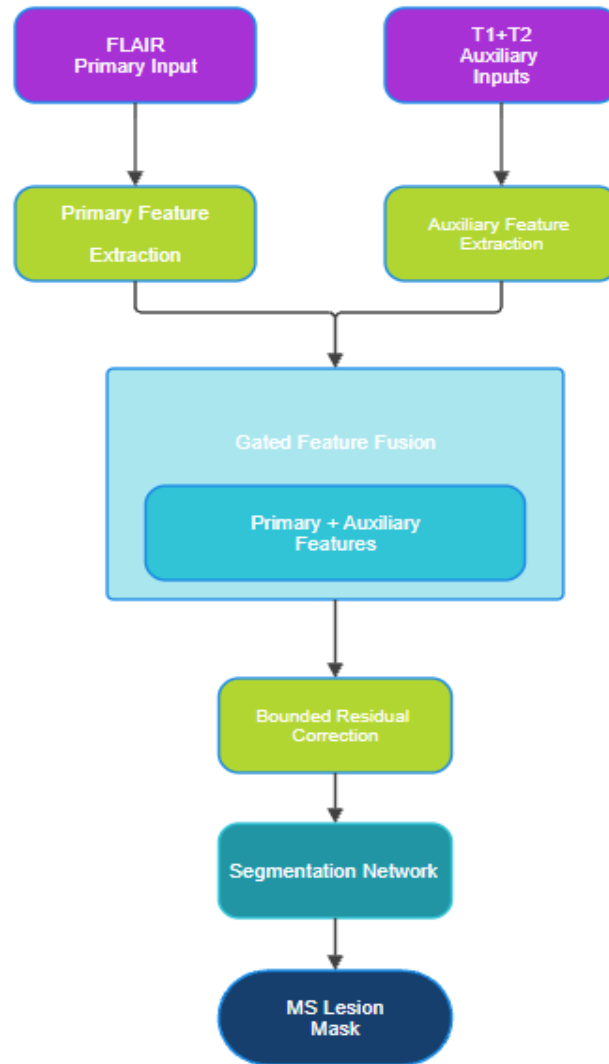


Figure 3.2.6 Proposed gated residual multimodal architecture for MS lesion segmentation.

3.2.7 Model Training and Optimization

Both the FLAIR-only UNETR baseline and the proposed Gated Residual Multimodal Network were trained using the same patient-level data partition and 3D training-sample generation strategy to maintain a consistent experimental setup. The models were trained on the preprocessed MRI data together with their corresponding binary ground-truth lesion masks. A supervised learning approach was used, where the MRI patches served as inputs and the corresponding lesion masks were used as target outputs. For the baseline, only FLAIR was provided as input. For the proposed model, FLAIR was used as the primary modality, while T1- and T2-weighted images were provided as auxiliary inputs.

Both models were optimized using Dice Cross-Entropy loss (DiceCELoss) with sigmoid activation. The Dice component encourages overlap between the predicted and ground-truth lesion regions, while the cross-entropy component provides voxel-level classification guidance. The loss configuration also included the squared prediction option to support segmentation optimization. During training, model parameters were updated iteratively through backpropagation according to the calculated segmentation loss. The validation set was evaluated throughout training to monitor segmentation performance, and the checkpoint with the highest validation Dice score was retained as the best-performing model. This checkpoint was then used for final testing instead of automatically selecting the model from the final training epoch. Training continued according to the predefined number of epochs, with validation performance monitored throughout the process. This allowed the model state with the strongest validation performance to be identified and avoided selecting a checkpoint solely based on its final training state.

After training, the selected checkpoint from each experiment was used for three-dimensional sliding-window inference on the held-out test patients. The resulting segmentation predictions were then compared with the corresponding ground-truth masks using the evaluation metrics described in the following methodology and results sections. The system workflow is showed in figure 3.2.7:

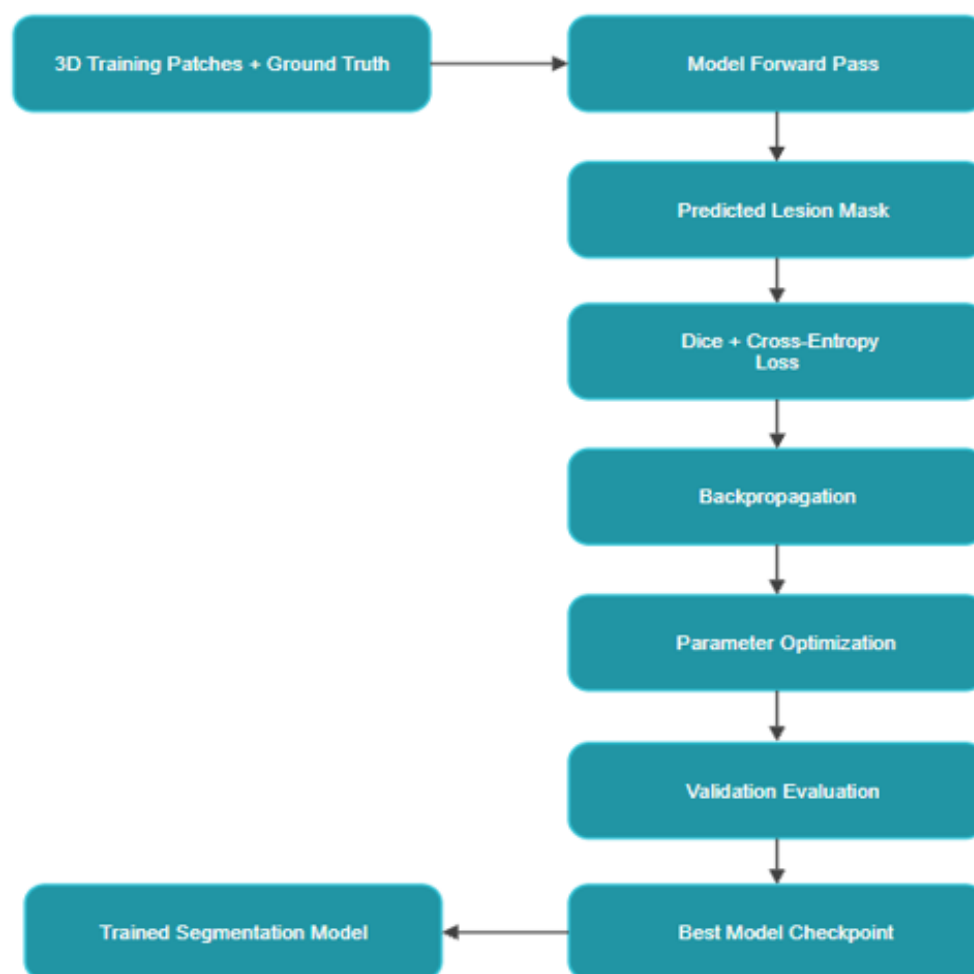


Figure 3.2.7: Model Training and Optimization

3.2.8 Three-Dimensional Sliding-Window Inference

After training, the best-performing checkpoint from each experiment was used to generate lesion segmentation predictions for the held-out test patients. Since the complete MRI volumes are three-dimensional and relatively large, three-dimensional sliding-window inference was used to process the full volumes in manageable regions. A fixed ROI size of $144 \times 144 \times 144$ voxels was used during inference. A 3D window was placed over a region of the MRI volume, and the corresponding region was passed through the trained model to generate a local lesion prediction. The window was then moved progressively across the volume, with overlapping regions used to ensure complete coverage. Predictions from the overlapping regions were combined to reconstruct a single whole-volume segmentation mask. This approach provides continuous coverage of the MRI volume while helping to reduce inconsistencies at the boundaries between individual regions.

The same inference procedure was applied to both experimental models. For the FLAIR-only UNETR, the preprocessed FLAIR volume was used as the model input. For the proposed multimodal model, the corresponding FLAIR, T1-weighted, and T2-weighted volumes were processed together according to the proposed architecture. The resulting whole-volume predictions were then compared with the corresponding ground-truth masks during evaluation.

Using a $144 \times 144 \times 144$ ROI allows the models to process the complete 3D MRI volumes in manageable sections rather than requiring the entire volume to be processed simultaneously. The reconstructed whole-volume segmentation masks were subsequently used for quantitative performance evaluation and lesion-level analysis.

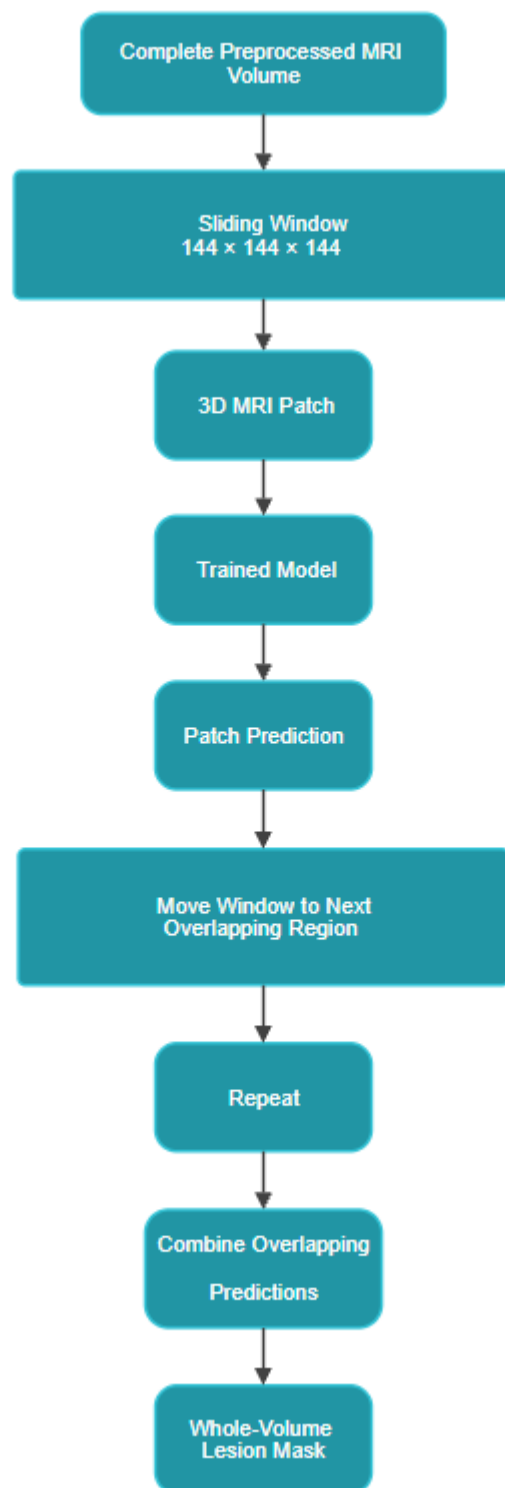


Figure 3.2.8: Three-dimensional sliding-window inference for whole-volume MS lesion segmentation.

3.2.9 Segmentation Performance Evaluation

The performance of the developed segmentation models was evaluated by comparing the predicted MS lesion masks with the corresponding ground-truth lesion masks from the held-out test patients. Both voxel-level segmentation metrics and lesion-level analysis were considered to provide a broader assessment of model performance.

Four commonly used segmentation metrics were selected: Dice Coefficient, Precision, Recall, and Intersection over Union (IoU). These metrics were calculated using the numbers of true-positive (TP), false-positive (FP), and false-negative (FN) voxels. True-positive voxels are lesion voxels correctly identified by the model, false-positive voxels are background voxels incorrectly classified as lesions, and false-negative voxels are ground-truth lesion voxels that were not detected.

The Dice Coefficient measures the spatial overlap between the predicted lesion mask and the ground-truth mask and is calculated as:

$$\text{Dice} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}} \dots\dots\dots \text{i}$$

A higher Dice value indicates greater agreement between the predicted and ground-truth lesion regions.

Precision measures the proportion of voxels predicted as lesions that are actually lesion voxels:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \dots\dots\dots \text{ii}$$

Higher precision indicates that the model produces fewer false-positive predictions.

Recall, also referred to as sensitivity, measures the proportion of ground-truth lesion voxels that are correctly detected:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \dots\dots\dots \text{iii}$$

Higher recall indicates that the model detects a larger proportion of the actual lesion regions.

Intersection over Union (IoU) measures the overlap between the predicted and ground-truth lesion masks relative to their union:

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \dots\dots\dots \text{iv}$$

For the overall test evaluation, these metrics were calculated from the predicted and ground-truth lesion voxels across the held-out test dataset. In addition to the overall or global measurements, patient-wise Dice scores were calculated to examine the consistency of segmentation performance across individual patients. This distinction is important because a global metric can be influenced more strongly by patients or lesions containing a larger number of lesion voxels.

The same evaluation procedure was applied to both the FLAIR-only UNETR baseline and the proposed Gated Residual Multimodal Network. The resulting metrics were used for quantitative comparison, while the performance of individual lesions and the effects of lesion size and FLAIR intensity were examined separately through lesion-level analysis.

3.2.10 Lesion-Level Analysis

To investigate the characteristics of individual Multiple Sclerosis (MS) lesions, a lesion-level analysis was performed using the ground-truth lesion masks from the training dataset. Unlike whole-volume evaluation, where all lesion voxels are considered together, lesion-level analysis treats each connected lesion as an individual observation. This provides a more detailed view of differences in lesion size and FLAIR intensity and provides a basis for examining segmentation performance under different lesion characteristics.

The analysis included 93 patient-timepoint cases and identified 2,529 individual lesions. Three-dimensional connected-component analysis was applied to the binary ground-truth masks using a 26-connectivity neighborhood. For each detected lesion, the number of lesion voxels was recorded as its size. The corresponding FLAIR image was then used to obtain lesion-wise intensity statistics, including the mean, median, minimum, and maximum FLAIR intensity.

Lesion Size and Intensity Characteristics

Lesion size was defined as the total number of voxels contained within each connected lesion. The analysis showed a highly variable and right-skewed distribution of lesion sizes. The size of the lesion was from 1 voxel to 67,929 voxels and the average was 447.43 with a median of 68 voxels. The difference between the mean and median is quite large, suggesting the presence of a few very large lesions that have a significant effect on the mean. The first Quartile (Q1) was 34 voxels and the third Quartile (Q3) was 153 voxels, which indicates that 50% of the detected lesions had between 34 and 153 voxels.

There was also significant variability in the mean FLAIR intensity in the lesion. It ranged from -0.1712 to 3.5822 , with a mean of 1.4907 and a median of 1.4629 . These values are not the actual MRI intensity but can be considered FLAIR Z-scores because they were normed using non-zero voxel Z-score during preprocessing. The mean FLAIR intensity of the first and third quartiles of the lesion was 1.1533 and 1.8302 , respectively.

The complete statistical distributions of lesion size and mean FLAIR intensity are presented in Table 3.2.10.

Table 3.2.10. Statistical Distribution of Lesion Size and Mean FLAIR Intensity

Statistic	Lesion Size (voxels)	Mean FLAIR Intensity (Z-score)
Number of lesions	2,529	2,529
Mean	447.43	1.4907
Standard deviation	2,473.95	0.4943
Min	1	-0.1712
Max	76929	3.5822

Lesion Size–Intensity Relationship

The voxel count for each lesion was plotted against the mean FLAIR Z score across all 2,529 lesions identified to further explore the correlation between lesion size and mean FLAIR Z score. The lesion-size axis was plotted on a logarithmic scale because the sizes of the lesions varied by several orders of magnitude from a single voxel to lesions that contain tens of thousands of voxels.

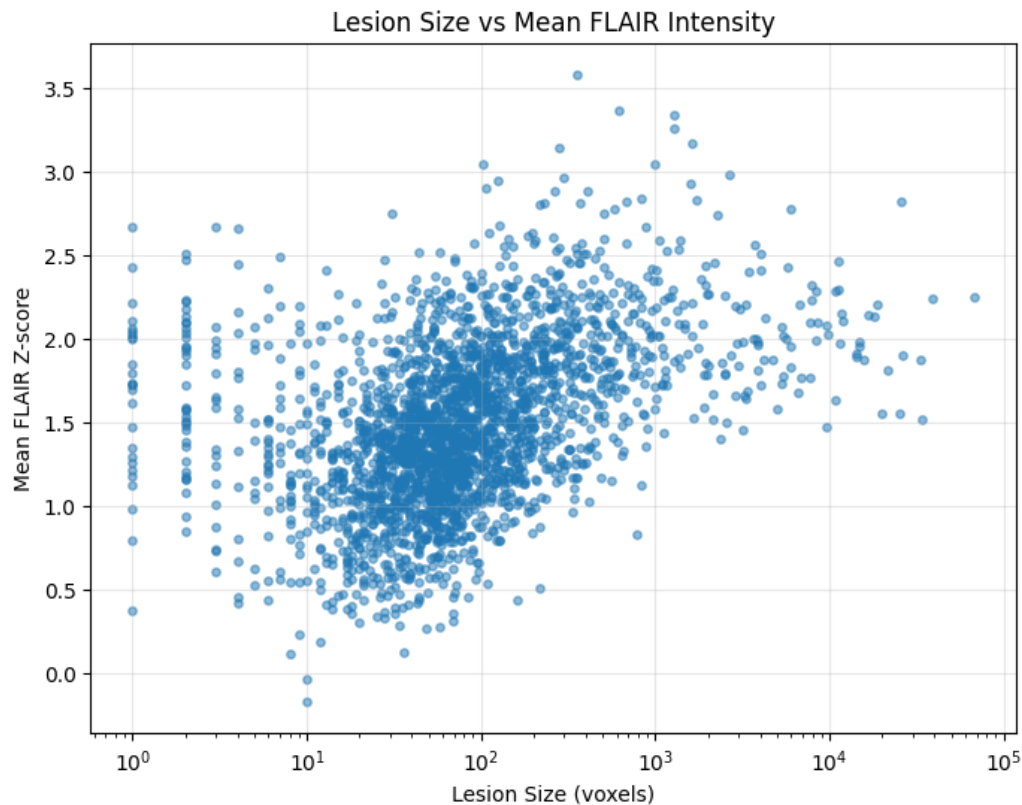


Figure 3.2.10: Lesion Size vs Mean Flair Intensity

The overall scatter graph illustrates a positive trend with increasing lesion size and mean FLAIR intensity, such that larger lesions appear to have higher mean FLAIR Z scores. However, the distribution of observations indicates that it is not unique. There is an obvious variability in the FLAIR intensity of lesions of similar size, and small and large lesions may be present over a wide window of normalized FLAIR intensity. Therefore, lesion size and FLAIR intensity are not equivalent properties for lesion difficulty; rather they are complementary.

The dense concentration of observations in the low size area also indicates a high level of unbalanced lesion sizes. The majority of lesions are small, and only a few large lesions are at the upper end of the size range. This imbalance is important for segmentation evaluation since the whole volume metrics could be greatly influenced by large lesions, while errors from small lesions might be relatively less on aggregate voxel-level metrics.

Lesion Classification for Subsequent Analysis

Data-driven thresholds were used based on the median lesion size and lesion mean FLAIR intensity values, for a systematic analysis of the lesion characteristics. The median lesion size was 68 voxels and the median mean FLAIR intensity of the lesions was 1.4629. These median values were then used to categorise the lesions according to their relative size and intensity of the lesions, as a lower/leaner or higher/fatter class without applying external thresholds.

The lesions were then divided into four groups according to the lesion size and the mean FLAIR intensity.

Table 3.2.11 Lesion Classification Based on Size and Mean FLAIR Intensity

Lesion Category	Number of Lesions	Lesions (%)	Total Lesion Voxels	Voxels (%)
Small + Non-bright	856	33.85	28770	2.54
Small + Bright	402	15.90	13309	1.18
Large + Non-bright	409	16.17	65236	5.77
Large + Bright	862	34.08	1024241	90.52
Total	2529	100.00	1131556	100.0

3.3 Project Plan

As the proposed MS lesion segmentation system is complex and a variety of activities are involved, a systematic project plan was prepared to organize the research and development process. The plan covered the major stages of the project, including literature review, data preparation, and preprocessing, model development, and training, evaluation, and system demonstration. To keep a clear workflow and to ensure that the various parts of the automated MS lesion segmentation system were developed in a structured way, each phase was broken down into specific tasks with defined outcomes.

The work flow for the overall project was split into the following phases:

Phase 1: Research and Requirement Analysis:

- i. Carried out a comprehensive literature review on automated Multiple Sclerosis lesion segmentation using MRI.
- ii. Reviewed existing deep learning-based segmentation methods, including U-Net-based architectures and Transformer-based models such as UNETR.
- iii. Examined the commonly used MRI modalities for MS lesion segmentation, with particular focus on FLAIR, T1-weighted, and T2-weighted images.
- iv. Identified major challenges in MS lesion segmentation, including small and low-intensity lesions, class imbalance, and variations in lesion characteristics.
- v. Defined the research objectives and established the experimental comparison between the FLAIR-only baseline model and the proposed multimodal approach.

Phase 2: Dataset Collection and Preparation:

- i. Collected and organized the MSLesSeg dataset, including FLAIR, T1-weighted, and T2-weighted MRI volumes along with their corresponding ground-truth lesion masks.
- ii. Structured the dataset according to patient and timepoint information to maintain consistent correspondence among the MRI modalities and lesion masks.
- iii. Applied a patient-level separation between the development data and the held-out test data to avoid patient overlap and reduce the risk of data leakage.
- iv. Checked the availability and correspondence of the required MRI modalities and ground-truth lesion masks for the selected cases.

Phase 3: MRI Preprocessing:

- i. Applied N4 bias-field correction to the MRI volumes to reduce low-frequency intensity non-uniformity.
- ii. Applied non-zero voxel Z-score normalization to standardize the MRI intensity distributions while keeping background voxels at zero.
- iii. Preserved the ground-truth lesion masks without modification during the intensity preprocessing steps.
- iv. Checked the consistency of image dimensions, modality correspondence, and preprocessing outputs.
- v. Performed quantitative and visual checks of the preprocessing results using intensity statistics, histograms, and representative MRI slices.

Phase 4: Baseline Model Development:

- i. Developed a FLAIR-only UNETR model as the baseline segmentation approach.
- ii. Used the preprocessed FLAIR volumes and corresponding ground-truth masks for supervised training.
- iii. Generated three-dimensional training samples using a patch size of $96 \times 96 \times 96$ voxels with lesion-aware positive and negative sampling.
- iv. Configured the UNETR architecture for three-dimensional MS lesion segmentation.
- v. Trained the baseline model and selected the best-performing checkpoint based on validation performance.

Phase 5: Proposed Multimodal Model Development:

- i. Designed the proposed Gated Residual Multimodal Network using FLAIR as the primary input and T1- and T2-weighted MRI as complementary inputs.
- ii. Developed separate primary and auxiliary branches to process the different MRI modalities.
- iii. Combined the multimodal features using Gated Feature Fusion to control the contribution of the auxiliary information.
- iv. Incorporated Bounded Residual Correction to refine the fused feature representation for lesion segmentation.
- v. Trained the proposed model using the same patient-level data separation and three-dimensional training strategy as the baseline model.

Phase 6: Model Training and Inference:

- i. Trained both the baseline and proposed models using a supervised deep learning approach.
- ii. Used DiceCELoss as the training loss function.
- iii. Monitored model performance throughout training and retained the checkpoint with the best validation performance.
- iv. Applied three-dimensional sliding-window inference to process complete MRI volumes.
- v. Used an inference region of interest (ROI) size of $144 \times 144 \times 144$ voxels to generate whole-volume lesion predictions.
- vi. Reconstructed the complete predicted lesion masks by combining the predictions from overlapping sliding windows.

Phase 7: Performance Evaluation and Comparative Analysis:

- i. Evaluated the predicted lesion masks against the ground-truth masks using the Dice coefficient, precision, recall, and Intersection over Union (IoU).
- ii. Compared the overall segmentation performance of the FLAIR-only UNETR baseline with that of the proposed Gated Residual Multimodal Network.
- iii. Performed lesion-level analysis to examine the segmentation behavior of individual lesions.
- iv. Classified the lesions according to size and mean FLAIR intensity using median-based thresholds.
- v. Analyzed four lesion categories: small/non-bright, small/bright, large/non-bright, and large/bright.

vi. Compared lesion detection between the baseline and proposed models to examine whether the complementary T1- and T2-weighted information helped recover difficult lesions.

Phase 8: System Demonstration and Documentation:

i. Developed a local web-based demonstration interface using Gradio to test the trained segmentation models.

ii. Designed the interface to accept FLAIR MRI as the primary input, with optional T1- and T2-weighted MRI inputs.

iii. I Set up the system to use baseline with FLAIR only when only FLAIR is supplied and the proposed multimodal model, with all three MRI modalities.

iv. Showed both the input MRI images and the lesion segmentation result predicted by the program on screen using the GUI.

v. Formulated methodology, experimental results, comparative analysis/future research that are documented in a final report.

Tools and Technologies Used:

i. Programming Language: Python 3.10

ii. Deep Learning Frameworks: PyTorch and MONAI

iii. Medical Image Processing: NiBabel and SimpleITK

iv. Numerical and Data Processing: NumPy, Pandas, and SciPy

v. Model Development: UNETR and the proposed Gated Residual Multimodal Network

vi. Training Hardware: NVIDIA GeForce RTX 3060 12 GB GPU

vii. Evaluation Metrics: Dice coefficient, Precision, Recall, and IoU

viii. Visualization: Matplotlib

ix. System Demonstration: Gradio

3.4 Task Allocation

The task allocation table shows the timeline of the major activities performed in the development of the automated Multiple Sclerosis lesion segmentation system. The project work was divided into various phases: literature survey, requirement analysis, dataset preparation, MRI preprocessing, baseline model building, proposed model building, model training and inference, performance evaluation, and system demonstration. The activities were done in parallel as well as sequential to optimize the development and training time.

Table 3.4: Task Allocation Table

Tasks	Weeks																		
	1 2	1 4	1 6	1 8	2 0	2 2	2 4	2 6	2 8	3 0	3 2	3 4	3 6	3 8	4 0	4 2	4 4	4 6	4 8
Data collection phase																			
Preproces s all the data																			
Model training																			
Create a demo Applicatio n.																			

The following table shows the chronology of the key activities undertaken during the project. The first phase was dedicated to literature review and requirement analysis to determine the direction of the research and to understand the problems of automated MS lesion segmentation. The MSLesSeg dataset was then collected and organized and the MRI preprocessing pipeline was implemented and verified. At the patient level, data splitting and three-dimensional training sample were then made for data preparation for model development.

The FLAIR-only UNETR model was developed first as the baseline, followed by the proposed Gated Residual Multimodal Network, which uses FLAIR as the primary modality and T1- and T2-weighted MRI as complementary inputs. Both models were subsequently trained and evaluated using three-dimensional sliding-window inference. Their predictions were assessed using voxel-level segmentation metrics and lesion-level comparative analysis. Finally, a Gradio-based demonstration system was developed, and the methodology, experimental findings, and overall project outcomes were documented in the final report.

3.5 Summary

This chapter described the research methodology, system design, and project planning strategy used to develop the automated Multiple Sclerosis (MS) lesion segmentation system. It covered the organization of the MSLesSeg dataset and the patient-level data separation used to maintain an appropriate experimental setup. The MRI preprocessing pipeline, including N4 bias-field correction and non-zero voxel Z-score normalization, was

also described, followed by three-dimensional training sample generation using lesion-aware sampling.

Two segmentation approaches were developed and described in this chapter. A FLAIR-only UNETR model was used as the baseline, while the proposed Gated Residual Multimodal Network used FLAIR as the primary modality and T1- and T2-weighted MRI as complementary inputs. The training and optimization procedures, three-dimensional sliding-window inference, and evaluation using Dice, precision, recall, and IoU were presented. The chapter also introduced the lesion-level analysis used to examine individual lesions based on their size and mean FLAIR intensity. Finally, the project plan and task allocation were presented to outline the scheduling and organization of the major project activities.

Chapter 4

Implementation and Results

This chapter describes the implementation and experimental analysis of the proposed automated Multiple Sclerosis (MS) lesion segmentation system using three-dimensional MRI data. It covers the development environment, dataset handling, MRI preprocessing, model training, three-dimensional inference, and performance evaluation procedures. The experimental results of the FLAIR-only UNETR baseline and the proposed Gated Residual Multimodal Network are also presented, followed by quantitative, lesion-level, and comparative analyses of their segmentation performance.

4.1 Environment Setup

Because the experiments were based on 3D MRI volumes and deep learning models, that are both very computation demanding, the environment configuration was an important part of the implementation of the proposed Multiple Sclerosis (MS) lesion segmentation system. All the development, preprocessing, training of the models, inference and evaluation processes were performed with local development tools and GPU computation. Local development, experimentation, data inspection, result analysis and verification of data preprocessing were carried out using Visual Studio Code and Jupyter Notebook under Windows 11 and Python 3.10.

The deep learning implementation was mainly based on PyTorch to train and test the model, and MONAI (Medical Open Network for AI) was used to process medical images and perform three-dimensional segmentation. The FLAIR-only UNETR baseline and the proposed Gated Residual Multimodal Network were implemented for 3D MRI segmentation. The computationally intensive training and inference procedures utilized an NVIDIA GeForce RTX 3060 GPU with 12 GB of VRAM. The installed PyTorch environment used PyTorch 2.5.1 with CUDA 12.4 support, while the NVIDIA driver environment reported CUDA 13.1 compatibility through NVIDIA-SMI.

For medical image handling, NiBabel was used to read and process NIfTI-format MRI volumes, while SimpleITK was used to perform N4 bias-field correction during preprocessing. NumPy was used for numerical operations and image-array manipulation. SciPy supported connected-component analysis in the lesion-level analysis, and Pandas was used to organize and analyze the experimental data. Matplotlib was used to generate MRI visualizations, intensity histograms, lesion-distribution plots, and other result figures.

The MSLesSeg dataset was organized into training and held-out test portions at the patient level. The training data consisted of patient-timepoint cases containing FLAIR, T1-weighted, and T2-weighted MRI volumes together with their corresponding ground-truth lesion masks. The preprocessing outputs were stored separately after N4 bias-field correction and non-zero voxel Z-score normalization. This organization allowed the preprocessing results to be verified independently before model training.

During model development, three-dimensional training samples of $96 \times 96 \times 96$ voxels were generated from the preprocessed MRI volumes. Lesion-aware sampling was used to increase the representation of lesion-containing regions during training. After training, complete MRI volumes were processed using three-dimensional sliding-window inference with an ROI size of $144 \times 144 \times 144$ voxels. The same inference strategy was applied to both the baseline and proposed models to maintain a consistent experimental setting.

For the final system demonstration, Gradio was used to develop a local web-based interface through which FLAIR, T1-weighted, and T2-weighted MRI volumes could be provided and the corresponding segmentation results visualized. The complete software environment and principal experimental configurations are summarized in Table 4.1.

The environment configuration was an important part of the implementation of the proposed Multiple Sclerosis (MS) lesion segmentation system because the experiments involved three-dimensional MRI volumes and computationally demanding deep learning models. The development, preprocessing, model training, inference, and evaluation procedures were carried out using local development tools together with GPU-based computation. Visual Studio Code and Jupyter Notebook were used for local development, experimentation, data inspection, preprocessing verification, and result analysis under Windows 11 with Python 3.10.

The deep learning implementation used PyTorch as the primary framework for training and inference, while MONAI was used for medical image processing and three-dimensional segmentation. The FLAIR-only UNETR baseline and the proposed Gated Residual Multimodal Network were implemented for 3D MRI segmentation. The computationally intensive training and inference procedures utilized an NVIDIA GeForce RTX 3060 GPU with 12 GB of VRAM. The installed PyTorch environment used PyTorch 2.5.1 with CUDA 12.4 support, while the NVIDIA driver environment reported CUDA 13.1 compatibility through NVIDIA-SMI.

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For the final system demonstration, Gradio was used to develop a local web-based interface through which FLAIR, T1-weighted, and T2-weighted MRI volumes could be provided and the corresponding segmentation results could be visualized. Table 4.1 gives an overview of the total software environment and important experimental conditions.

Table 4.1: Training Environment and Configuration Summary

Component	Details
Operating System	Windows 11
Programming Language	Python 3.10
Deep Learning Framework	PyTorch 2.5.1 + CUDA 12.4
Medical AI Framework	MONAI
Baseline Model	UNETR
Proposed Model	Gated Residual Multimodal Network
GPU	NVIDIA GeForce RTX 3060, 12 GB VRAM
Development Environment	Visual Studio Code, Jupyter Notebook
Medical Image Format	NIfTI (.nii.gz)
Medical Image Processing	NiBabel, SimpleITK
Numerical Processing	NumPy, SciPy
Data Analysis	Pandas
Visualization	Matplotlib
MRI Modalities	FLAIR, T1-weighted, T2-weighted
Training Patch Size	$96 \times 96 \times 96$ voxels
Inference ROI Size	$144 \times 144 \times 144$ voxels
Processing	N4 bias-field correction and non-zero voxel Zscore normalization
Segmentation Loss	DiceCELoss
Evaluation Matrices	Dice, Precision, Recall, IoU
System Demonstration	Gradio

4.2 Testing and Evaluation/Performance/ Comparative Analysis

After training, the developed models were tested thoroughly with an unseen test set for evaluating the generalization and segmentation performance. The test set consisted of 22 patients that were not included in the MSLesSeg data set (P54–P75). To prevent overlap and/or information leakage from the training data to the test data, patient-level separation was maintained throughout the experimental setup. Three-dimensional MRI volumes were used for all models to be assessed with the same unseen patients.

To evaluate various segmentation models, the following models were investigated: 3D U-Net, 3D Attention U-Net, UNETR, and the proposed Gated Residual Multimodal Network. The input modality for the 3D U-Net, 3D Attention U-Net, and UNETR models was FLAIR MRI while the proposed model was multimodal with FLAIR, T1-weighted and T2-weighted MRI. Precision, recall, Dice coefficient and Intersection over Union (IoU) were used for the evaluation of the models. The Dice coefficient was used to compute the spatial overlap between the ground-truth lesion masks and the predicted masks. The precision was defined as the ratio of the number of correctly classified lesion voxels to the total number of predicted lesion voxels, and the recall was defined as the ratio of the number of voxels classified by the model as lesion to the number of voxels that were truly lesions. IoU achieved an extra assessment of the overlap between the predicted and groundtruth masks.

UNETR obtained the best global Dice score of 71.23%, the best global precision of 74.72% and the best global IoU of 55.31% among the evaluated models. The 3D U-Net gave a global Dice score of 70.21% and the 3D Attention U-Net got 67.59%. The Dice score of the proposed Gated Residual Multimodal Network was 70.09% globally. While it did not have the best global Dice score, the proposed model had the best global recall, which is 74.06% compared with 68.72% for 3D U-Net, 63.00% for 3D Attention U-Net, and 68.05% for UNETR. This means that the voxels that were predicted by the proposed approach had a higher proportion of the ground-truth lesion voxels. However, its accuracy on a global scale was lower at 66.54%, which indicates that along with an increase in the detection of the

Table 4.2: Comparative Performance of the Evaluated Models

Model	Input Modality	Global Precision	Global Recall	Global Dice	Global IoU
3D U-Net	FLAIR	71.76%	68.72%	70.21%	54.09%
3D Attention U-Net	FLAIR	72.91%	63.00%	67.59%	51.04%
UNETR	FLAIR	74.72%	68.05%	71.23%	55.31%
Proposed Model	FLAIR + T1 + T2	66.54%	74.06%	70.09%	53.96%

4.3 Results and Discussion

4.3.1 Voxel-Level Results

The voxel-level evaluation was conducted to evaluate and compare the overall segmentation performance of the four models on the same unseen test set, P54–P75 (22 patients). Global Dice coefficient, precision, recall and Intersection over Union (IoU), mean patient-wise dice and its standard deviation were included in the assessment. These metrics are combined to give a complete picture of each model's ability to identify MS lesion voxels compared with the corresponding ground-truth masks.

Table 4.3 shows the voxel-level results of the four models evaluated. There are noticeable differences between the overall performance of the models in terms of segmentation. UNETR obtained the best global Dice score of 71.23% as well as the best precision of 74.72% and IoU of 55.31%. Conversely, the proposed model achieved the highest recall of 74.06%, indicating its better performance in detecting the lesion voxels in the ground-truth masks. The 3D U-Net scored 70.21% Dice on a global level while the 3D Attention U-Net scored 67.59% Dice on a global level.

To ease the interpretation of the comparison, the global segmentation metrics of the four models are represented as a bar chart. The chart displays the difference between the models for Dice, precision, recall and IoU.

Voxel-Level Performance Comparison: Comparison of global Dice, precision, recall, and IoU among the four evaluated MS lesion segmentation models.

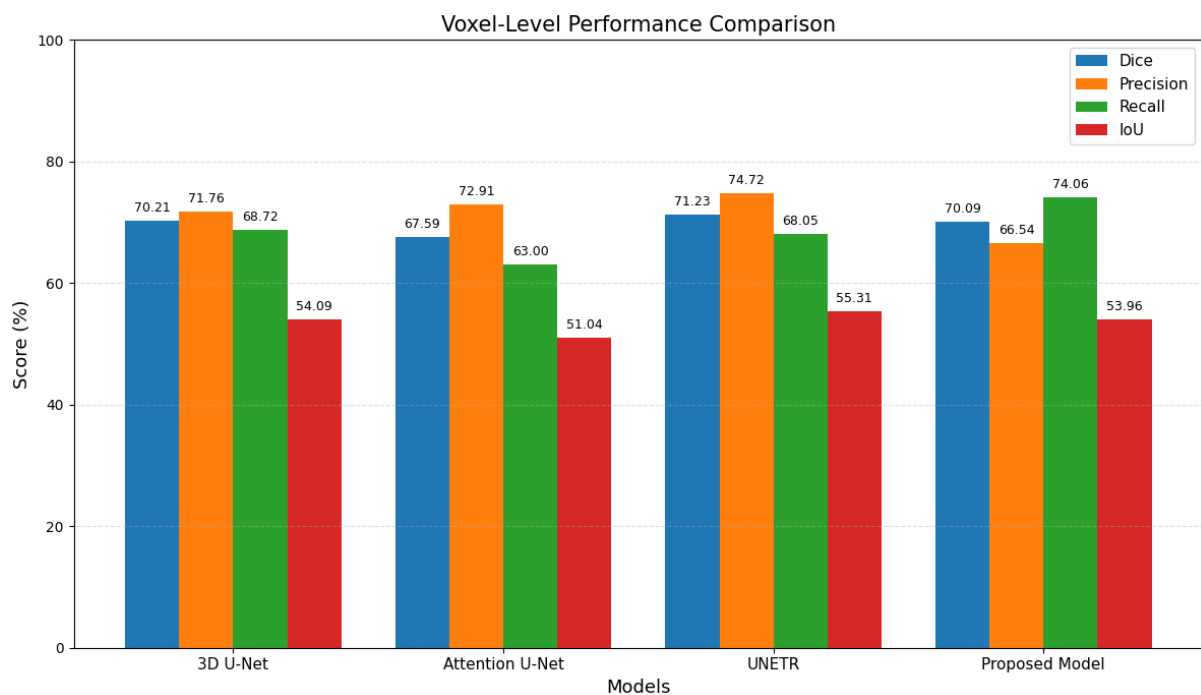


Figure 4.3.1: Comparative global voxel-level performance of the evaluated models

The bar chart indicates that UNETR achieves the strongest overall performance in terms of Dice, precision, and IoU. In comparison, the proposed model records the highest recall among the evaluated approaches. This suggests that the proposed model is able to detect a larger number of lesion voxels. However, its comparatively lower precision indicates that this higher detection rate also results in a greater number of false-positive predictions.

Box Plot for Per-Patient Dice

To assess the consistency of segmentation performance across individual patients, the patient-wise Dice scores were analyzed for all 22 patients in the test set. Unlike the global Dice score, which is calculated by combining lesion voxels from the entire test set, the patient-wise Dice score is calculated separately for each patient. As a result, the distribution of patient-wise Dice scores provides a better understanding of how the models perform across different individual cases and how much their performance varies from patient to patient.

The distribution of patient-wise Dice scores for the four models is shown with a box plot. The plot displays the median, interquartile range and the distribution of scores, indicating the overall range and the extreme scores at the patient level. This facilitates comparison of the models consistency within each test case.

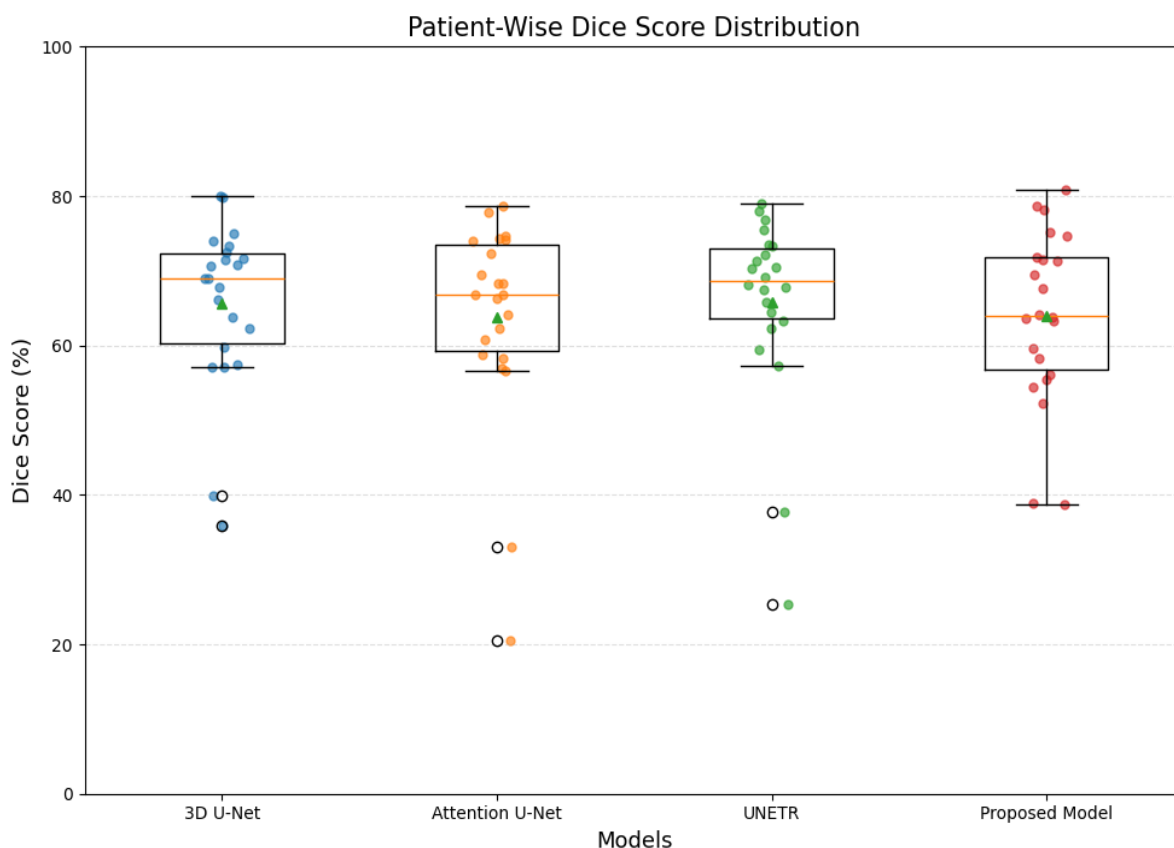


Figure 4.3.2: Distribution of patient-wise Dice scores for the four evaluated models.

The results for the patient-wise analysis show that the segmentation performance is different for the 22 test patients for all the four models. UNETR obtained the best mean patient-wise Dice score of 65.85%, which was very close to the score of 3D U-Net (65.64%). The mean patient-wise Dice score of the proposed model was 64.00% as compared to 63.77% for Attention U-Net.

The standard deviation values also show differences in performance consistency across patients. Attention U-Net exhibited the highest variation, with a standard deviation of 13.82%, whereas the proposed model had a lower standard deviation of 11.35%. Although the proposed model did not achieve the highest mean patient-wise Dice score, its lower standard deviation suggests that its performance remained relatively consistent across the individual test patients.

4.3.2 Statistical Significance Testing

To determine whether the differences in patient-wise segmentation performance were statistically significant, a paired statistical analysis was conducted using the Dice scores from the same 22 test patients. Because the patient-wise Dice scores may not follow a normal distribution and the measurements are paired across the different models, the Wilcoxon signed-rank test was selected to compare the proposed model with each baseline model. Along with the p-value, rank-biserial correlation was also calculated as an effect-size measure to assess the magnitude and direction of the differences between the models.

Table 4.3.2: Statistical Significance Testing

Comparison	N	Wilcoxon W	P-value	Rank Biserial Correlation
Proposed vs 3D U-Net	22	106.0	0.5235	-0.1621
Proposed vs Attention U-Net	22	104.0	0.4826	+0.1779
Proposed vs UNETR	22	96.0	0.3369	-0.2411

The results of the Wilcoxon signed-rank test showed that none of the differences between the proposed model and the baseline models were statistically significant at the 0.05 significance level. The comparison between the proposed model and 3D U-Net resulted in a p-value of 0.5235, while the comparisons with Attention U-Net and UNETR produced p-values of 0.4826 and 0.3369, respectively. The corresponding rank-biserial correlation values were -0.1621, 0.1779, and -0.2411.

These findings suggest that, although differences in patient-wise Dice performance were observed among the models, the results from the 22-patient test set do not provide sufficient statistical evidence to conclude that the proposed model performs significantly differently from any of the baseline models.

4.3.3 Size × Brightness Groups

To examine segmentation performance at the individual-lesion level, 874 matched lesions from the UNETR and Phase 2 predictions were analyzed based on lesion size and mean FLAIR intensity. Using the median lesion volume (63 voxels) and median mean FLAIR intensity (1.4307) as thresholds, the lesions were separated into four groups.

Lesions with <63 voxels were considered small, and those with ≥63 voxels were considered large. The intensity of FLAIR signal was rated as low intensity if the mean intensity of the lesion was below 1.4307, and high intensity if the mean intensity of the lesion was 1.4307 or more. Based on these two characteristics, the lesions were grouped into four categories: small + low, small + high, large + low, and large + high. All categories results are added in figure 4.3.3

Table 4.3.3: Lesion-Level Performance According to Size and FLAIR Intensity

Difficulty Group	Lesions	UNETR Mean Recall	Phase 2 Mean Recall	Recall Improvement	Rescued	UNETR Only	Misclassified By Both	Rescued By Both
Small + Low	310	0.1415	0.2065	+0.0650	30	7	177	96
Small + High	126	0.5348	0.5002	-0.0346	8	20	18	80
Large + Low	127	0.2661	0.3915	+0.1254	34	2	18	73
Large + High	311	0.7140	0.7891	+0.0750	9	4	2	296

The results indicate that both lesion size and intensity influenced lesion-level recall. The most noticeable improvement with Phase 2 was observed in the large + low-intensity group, where recall increased from 0.2661 with UNETR to 0.3915 with Phase 2, corresponding to an improvement of 0.1254. Phase 2 also showed improved recall for the small + low and large + high groups. However, for small + high lesions, UNETR achieved a higher recall than Phase 2, resulting in a decrease of 0.0346 for Phase 2.

Among the four groups, large + high lesions achieved the highest recall for both models. This suggests that larger lesions with relatively high FLAIR intensity were easier for both approaches to detect compared with the other lesion groups.

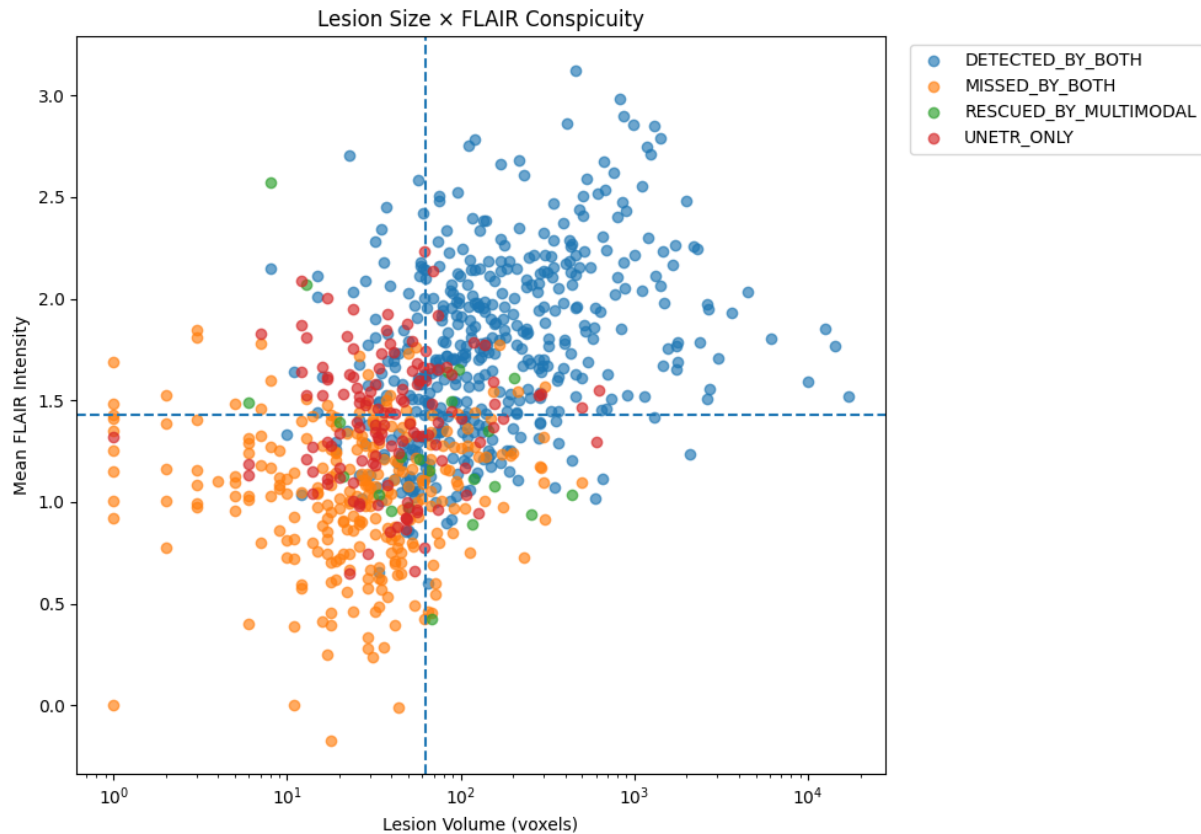


Figure 4.3.3: Lesion size vs Flair Conspicuity

4.3.3 Recall Comparison

The mean recall was computed for all 874 matched lesions for a direct comparison between the lesion detection capability of UNETR and the proposed Phase 2 model. In contrast to voxel-level recall, which reflects the number of correctly detected lesion voxels, lesion-level recall reflects the amount of each individual lesion correctly detected. This offers a more fine-grained insight into the ability of the two models to recognize individual MS lesions.

Table 4.3.3: Lesion-Level Recall and Detection Rate Comparison

Metric	UNETR	Phase 2	Improvement
Mean Lesion Recall	0.4200	0.4830	+0.0630
Lesion Detection Rate	0.6613	0.7162	+0.0549

The mean lesion recall of Phase 2 was 0.4830, which is higher than the 0.4200 mean lesion recall for UNETR, thus improving by 0.0630. The lesion detection rate also improved from 66.13% using UNETR to 71.62% using Phase 2, which is an improvement of 5.49 percentage points.

Results showed that the proposed Phase 2 model outperformed the UNETR baseline model in detecting individual MS lesions. This improvement was seen despite the difference in overall voxels level segmentation performance between the two models.

4.3.3 Brightness-Decile Analysis

For a more detailed analysis of the effect of lesion intensity on segmentation performance, the 874 lesions were split into ten brightness deciles based on the mean FLAIR intensity. About 10% of the lesions were in each decile. Then the recall was separately computed for UNETR and proposed model in each decile.

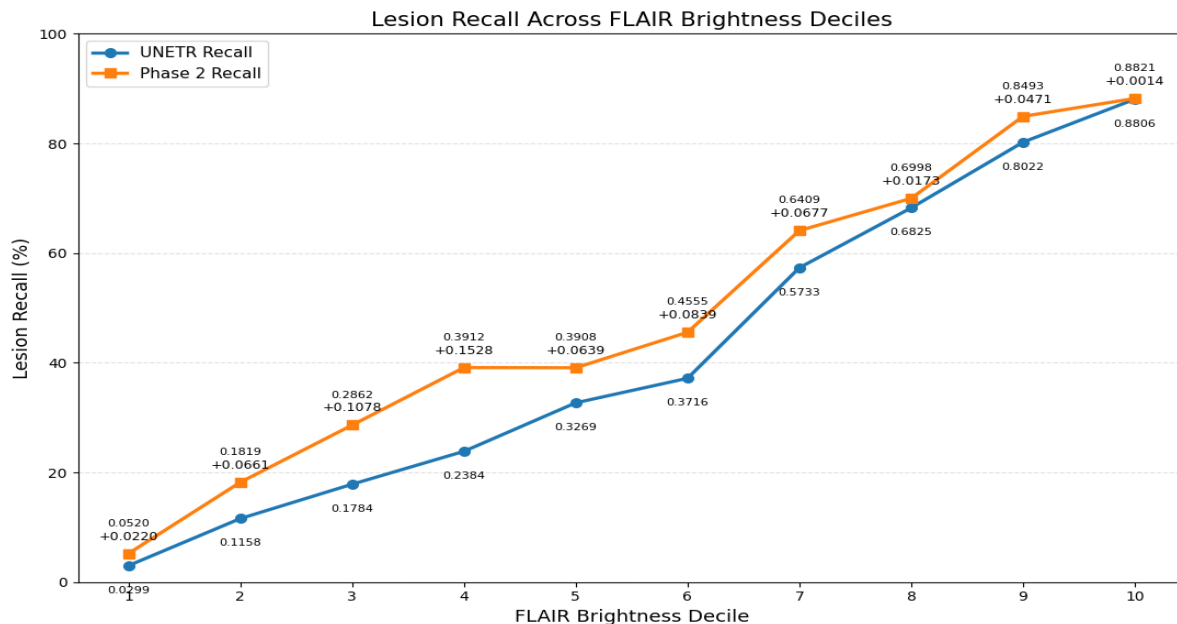


Figure 4.3.3: Lesion Recall across FLAIR Brightness Deciles

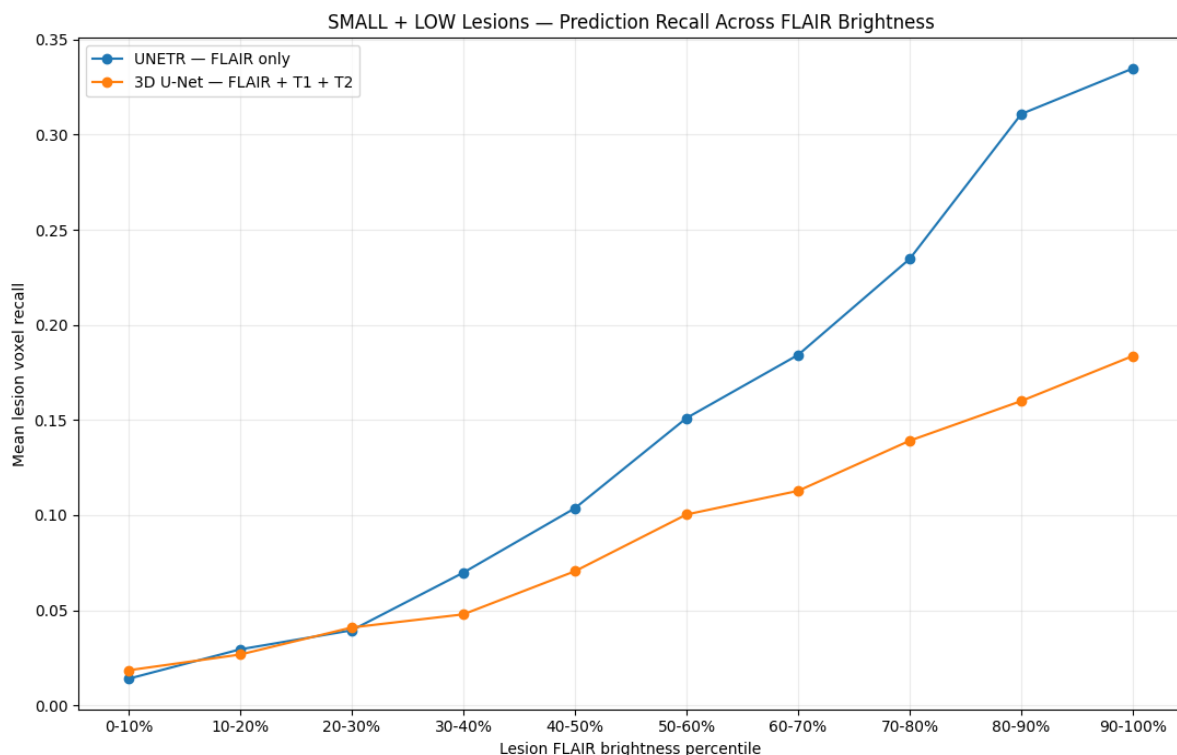


Figure 4.3.3: Prediction recall across Flair Brightness

These brightness-decile analyses show a good correlation between lesion intensity and detection performance for both models. For UNETR, recall went up from 0.0299 for the darkest decile to 0.8806 for the brightest decile. The same trend was noticed for Phase 2, where there was an increase in recall from 0.0520 to 0.8821. Overall, Phase 2 had higher recall than UNETR at all brightness deciles.

The largest improvement was observed in Decile 4, where Phase 2 increased recall by 0.1528 compared with UNETR. In contrast, the improvement was minimal in the highest-intensity decile, where both models already achieved relatively high recall. This suggests that the advantage of Phase 2 was more noticeable for lesions with lower or moderate intensity, while the performance difference became smaller for the brightest lesions.

4.3.3 Rescue/Harm Analysis

A rescue and harm analysis was conducted using the 874 exactly matched lesions to examine how Phase 2 affected individual-lesion detection compared with UNETR. A lesion was classified as rescued when it was missed by UNETR but successfully detected by Phase 2. In contrast, a lesion was classified as UNETR-only when it was detected by UNETR but missed by Phase 2. Lesions detected by both models and those missed by both models were also included in the analysis to provide a complete comparison of their lesion-level detection performance.

Table 4.3.3: Lesion-Level Rescue/Harm Analysis

Lesion Status	Number of Lesions	Percentage
Detected by Both	545	62.36%
Rescued by Phase 2	81	9.27%
UNETR Only	33	3.78%
Missed by Both	215	24.60%
Total	874	100%

Among the 874 matched lesions, 545 lesions (62.36%) were detected by both models. Phase 2 successfully rescued 81 lesions (9.27%) that were missed by UNETR, while 33 lesions (3.78%) were detected only by UNETR and missed by Phase 2. This indicates that Phase 2 recovered more individual lesions than it lost compared with the UNETR baseline. A total of 215 lesions (24.60%) were missed by both models. Overall, Phase 2 produced a net gain of 48 detected lesions, resulting in a rescue-to-harm ratio of 2.4545. These findings suggest that the proposed Phase 2 model achieved more lesion-level recoveries than losses when compared with UNETR.

4.3.4 Qualitative Results

In addition to quantitative evaluation, a qualitative evaluation was performed to visually analyze the segmentation behavior of the UNETR model on representative test cases. Three cases were chosen to reflect varying degrees of segmentation performance: a high performing case, a case that was in the middle, and a low performing case. In each case, the FLAIR MRI image, the ground-truth lesion mask, the predicted segmentation and the overlay of the ground-truth and predicted masks are shown, with the corresponding slice-level Dice score. Additional visualizations of the ground-truth and predicted masks,

together with the TP/FP/FN map, are included to provide a more detailed view of the agreement between the predicted and reference segmentations and to highlight the regions where segmentation errors occurred.

High-Performing Case

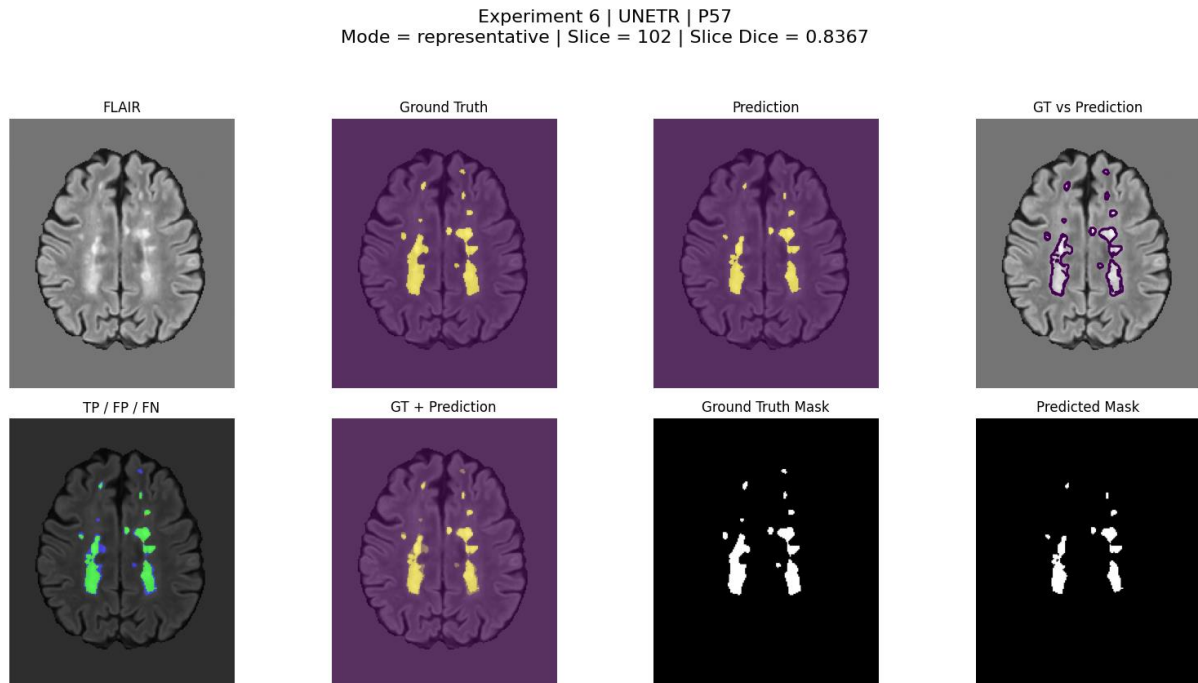


Figure 4.3.4: Qualitative segmentation result of UNETR for P57 (Slice 102, Dice = 0.8367)

Figure 4.3.4 shows a representative high-performing case for the UNETR model, obtained from patient P57 at slice 102. The predicted lesion regions show a high level of spatial agreement with the ground-truth segmentation, particularly in the larger and more clearly defined lesion areas. The overlay of the ground-truth and predicted masks further shows that most of the predicted lesion boundaries closely match the reference segmentation. The corresponding slice-level Dice score of 0.8367 indicates a strong overlap between the predicted and ground-truth lesion regions in this representative slice.

Median/Representative Case:

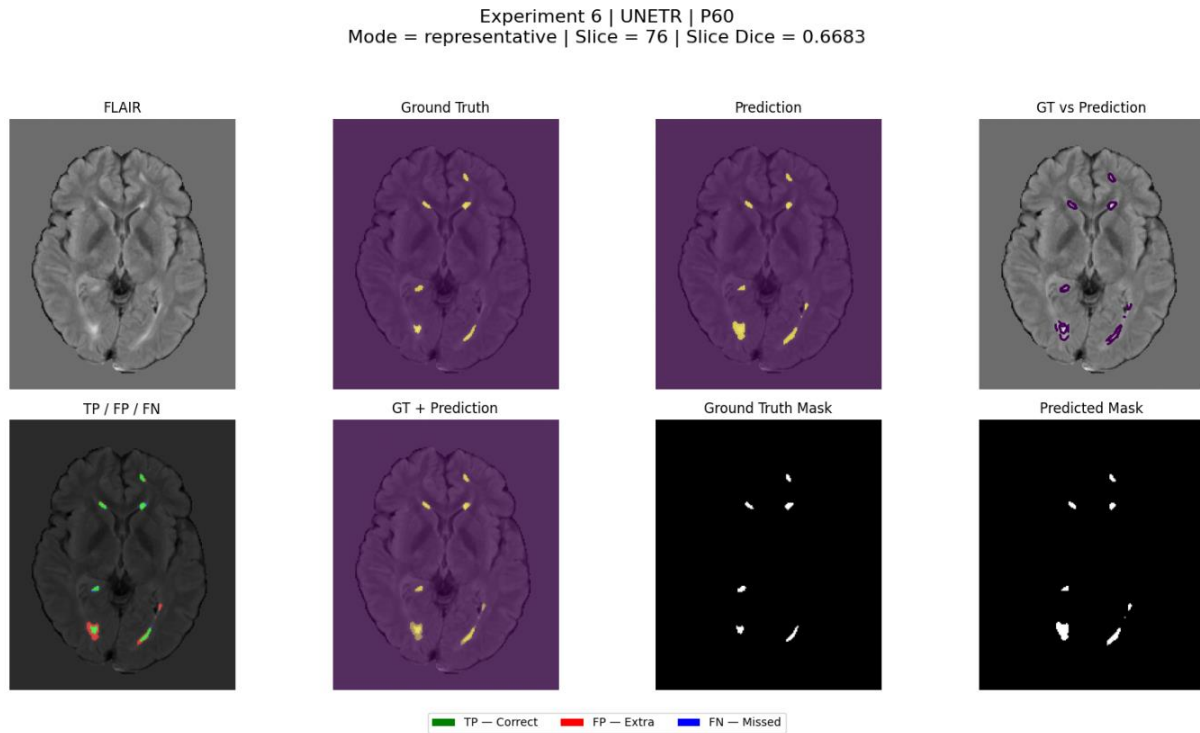


Figure 4.3.5: Qualitative segmentation result of UNETR for P60 (Slice 76, Dice = 0.6683).

Figure 4.3.5 presents a representative intermediate-performance case for the UNETR model from patient P60 at slice 76. The prediction successfully captures several of the lesion regions visible in the ground-truth mask, although some differences in lesion size and extent can be observed. There are some other predicted areas that are not related to the reference lesions. The TP/FP/FN visualization helps to better understand these differences by showing the areas in which the prediction matches the ground truth, and the areas in which the model made extra predictions or missed to fully capture the reference lesions. The Dice score obtained at the slice level is 0.6683 which shows the moderate spatial agreement between the predicted lesion region and ground truth lesion region.

Low-Performing Case:

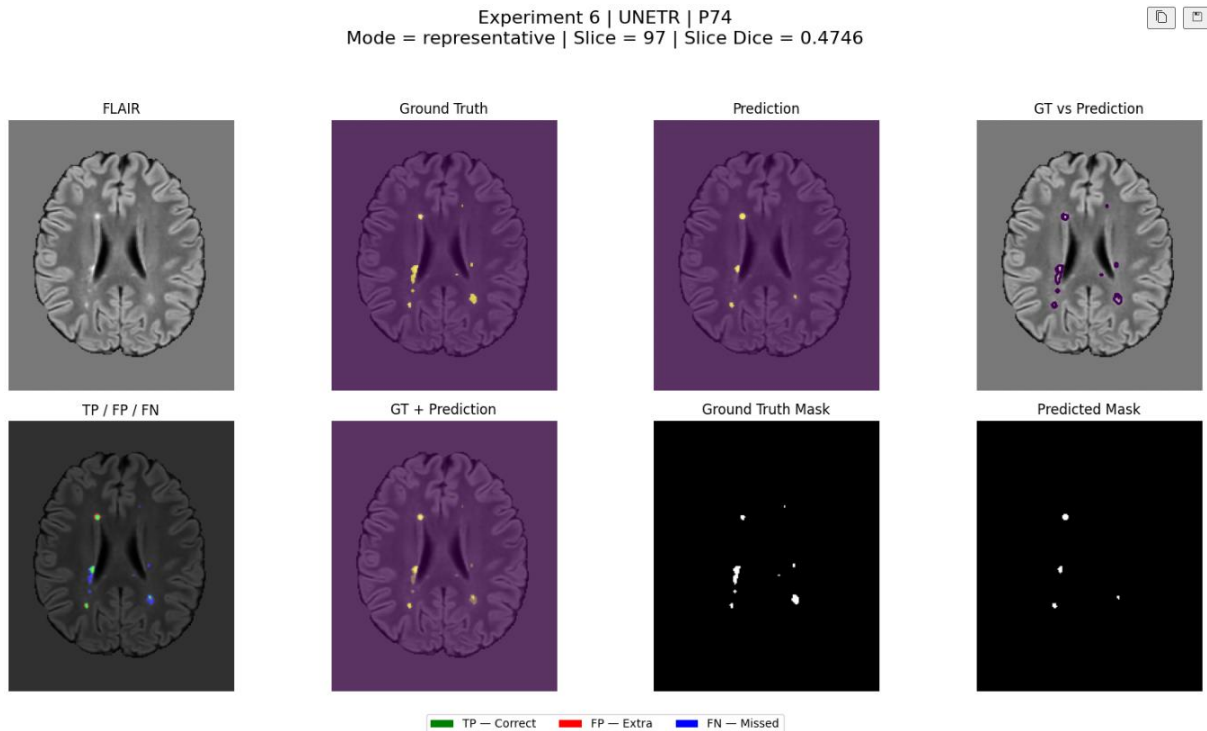


Figure 4.3.6: Qualitative segmentation result of UNETR for P74 (Slice 97, Dice = 0.4746)

Figure 4.3.6 shows an example of a poor performance case for the UNETR model from patient P74 at slice 97. There is a larger discrepancy between the ground truth lesion area and the prediction in comparison with the high performing case. The model only partially covers several lesion areas, and there are also areas that are predicted that are visible in the TP/FP/FN visualization. These differences result in a lower Dice score for the slice-levels (0.4746) reflecting lower spatial agreement between the prediction and the ground truth. The result also highlights the difficulty of accurately segmenting lesions in this particular case.

Overall Qualitative Discussion

Qualitative examples demonstrate that when the lesion regions are well defined and sufficiently covered by the slice being used, UNETR works well. However, its segmentation performance decreases in more challenging cases. It is also evident in the comparison of the high, intermediate, and low-performing examples that the same model can yield significantly different results with different characteristics, size, and spatial distribution of the lesions. These qualitative observations are in line with the patient-wise quantitative variation. The visual results further illustrate its capability to correctly segment lesion areas in good cases and the segmentation errors in more challenging cases.

4.4 Summary

In this chapter, the implementation and experimental assessment of the proposed automated Multiple Sclerosis lesion segmentation system was presented. The entire pipeline consisted of preparing MRI data, N4 bias-field correction, non-zero voxel Z-score normalization, model training, and three-dimensional sliding-window inference on held-out test patients. Four Deep learning models were implemented and evaluated, 3D U-Net, 3D Attention U-Net, UNETR and proposed Phase 2 gated residual multimodal model. The proposed model used FLAIR, T1-weighted, and T2-weighted MRI as the primary input via gated feature fusion and residual correction, while the first three models used FLAIR MRI as the primary input. The experimental outcomes revealed that all the four models were able to detect and segment the MS lesions, but there were differences in the four models based on the evaluation measures. Global and patient-wise quantitative assessment was conducted by the Dice coefficient, precision, recall, and IoU. As shown in the comparative results, the proposed Phase 2 model got the best global recall of 0.7406 which means that it was more sensitive towards the lesion regions, while UNETR got the best Dice score of 0.7123 globally. Moreover, qualitative analysis was conducted on representative high, intermediate and low-performing cases to visually assess the results of the segmentation.

The results shown in this chapter validate the experimental performance of the models developed in this chapter and offer quantitative and qualitative comparisons between the proposed multimodal approach and existing three-dimensional segmentation architectures.

Chapter 5

Engineering Standards and Design Challenges

The engineering standards, development practices and important challenges that were taken into account during the design and development of the automated Multiple Sclerosis (MS) lesion segmentation system are discussed in this chapter. It includes information about the software and hardware used in the project, as well as the way that data is handled and the way that communication takes place during the project. The chapter also discusses the social, environmental, ethical and sustainability issues associated with the proposed system. The project is also analyzed from the point of view of complex engineering problem solving and the engineering activities in the development, implementation, and evaluation of the deep learning-based medical image segmentation system.

5.1 Compliance with the Standards

The software engineering, medical image processing, machine learning and research methodology standards and practices were used for the development of the proposed MS lesion segmentation system. These practices were applied for ensuring reliability, reproducibility, maintainability and responsible use of the developed system. Of particular interest was the consistency in the data preprocessing, the use of structured approaches to model development and performance evaluation, and software implementation practices. The project also addressed the ethical and responsible use of medical imaging data, ensuring that it is handled in a responsible manner to facilitate ethical and trustworthy research.

5.1.1 Software Standards

The proposed MS lesion segmentation system was implemented in a standard software engineering manner and deep learning practices to ensure maintainability, reproducibility, and systematic experimentation. The implementation was mostly done in Python, utilizing widely used science computing and deep learning libraries such as PyTorch, MONAI, NumPy, NiBabel, and SimpleITK. The whole procedure was divided into several stages: data preparation, data preprocessing, model training, inference and performance evaluating. Standardized methods were used to process the MRI data (N4 bias-field correction and non-zero voxel Z-score normalisation). To obtain consistency along the experimental pipeline, both training and test data were subjected to the same pre-processing steps. Patient-level separation of the data was also performed to minimize the risk of data leakage between the training, validation and testing sets. As the baseline models, a series of 3D segmentation architectures were built: 3D U-Net, 3D Attention U-Net, and UNETR. The proposed model is multimodal and integrates the 3 modalities

(FLAIR, T1 weighted, and T2 weighted MRI) via gated feature fusion and residual correction. The models' performance was assessed by widely used segmentation metrics such as Dice coefficient, precision, recall, and Intersection over Union (IoU). All the trained models were saved in PyTorch checkpoint files, allowing for future inferences and also reproducibility without retraining the model. Complete MRI volumes were processed using 3-dimensional sliding-window inference without consuming too much GPU memory. GPU-based computation has also been used during training and inference to boost processing volumetric medical images.

Overall, these software engineering practices proved beneficial in structuring and reproducing the developing, testing and comparing the MS lesion segmentation models. This also helped to have uniformity of the stages in the experimental workflow and to evaluation of the proposed system.

5.1.2 Hardware Standards

The proposed MS lesion segmentation system was designed considering the computational demands of 3D medical image processing and model training in deep learning. Development and tests were done in a GPU enabled computing environment to efficiently process the volumetric MRI data. The model training and the model inference were performed on an NVIDIA GeForce RTX 3060 GPU with a VRAM of 12 GB, which was sufficient for the 3D segmentation architectures used.

GPU acceleration was achieved using CUDA and PyTorch, to minimize training time and time taken for three dimensional sliding window inference. GPU-based computation was very significant as the models used were based on volumetric MRI data that are memory and computing resource intensive. A memory-efficient inference technique was also employed to handle the processing of entire MRI volumes within the limited GPU memory.

The hardware used was selected based on the computing needs for the project and not necessarily medical imaging hardware. Consequently, the created system can be applied to other computing platforms with similar GPU setup making it feasible for research applications and as a starting point to be deployed in the future.

5.1.3 Communication Standards

Communication and documentation practices were well maintained throughout the development of the MS lesion segmentation system, so as to facilitate systematic development and keep a record of research activities. Structured file/directory organization with consistent naming conventions was used to organize project-related source code, experimental scripts, model checkpoints, preprocessing procedures and results. To handle multiple experimental configuration and model versions (including 3D U-Net, 3D Attention U-Net, UNETR, and the proposed Phase 2 multimodal model), version control and documentation practices have been used. The training configurations, validation results, checkpoints of models, predictions results and evaluation results were documented to facilitate the comparison of the various approaches and the reproduction of the experiments.

Regular communication and reviews of experimental results were also used for communication and progress monitoring during the project. During the development

process important decisions were recorded for processing, structure of the model, training process and evaluation methods. Such practices were used to keep experiments standardized, coordinated and traceable and reproducible throughout the various phases of the research project.

5.2 Impact on Society, Environment and Sustainability

The proposed automated Multiple Sclerosis (MS) lesion segmentation system can be applied to the fields of healthcare, medical research and computer-assisted medical image analysis. The system can automatically detect and segment the lesion regions in three dimensional MRI images, which can help researchers and healthcare professionals to analyze the location and size of MS lesions. This section covers the possible impact of the designed system on human lives, society and environment, as well as ethical and long-term sustainability issues.

5.2.1 Impact on Life

To facilitate the MRI analysis, an automated MS lesion segmentation system can be used to reduce manual time and effort spent on the identification and delineation of lesion regions. Manual segmentation is time consuming and can result in inter-observer variation. An automated method may be able to achieve more uniform segmentation results, thereby helping clinical and research personnel in the analysis of medical images. The system built handles the 3D MRI data and generates lesion segmentation masks which can be quantitatively evaluated using various metrics like Dice coefficient, precision, recall, IoU etc. Such automated analysis could be useful in quantifying the amount of lesions and could aid further research into abnormalities in the brain associated with MS.

In fact, the proposed system is conceived as a computer-aided tool to be used in the research field and not as a stand-alone clinical diagnostic system. Before using the predictions generated to inform clinical decision making, the predictions must therefore be reviewed and interpreted by qualified medical professionals.

5.2.2 Impact on Society & Environment

The proposed system can help in the increasing use of AI in medical image analysis, especially in the research of neurological diseases like Multiple Sclerosis. Automated lesion segmentation can be useful for researchers to process larger collections of MRI scans more efficiently, and can be useful for the development of computer-assisted medical imaging technologies. Multimodal MRI analysis (combining information from FLAIR, T1-weighted and T2-weighted images) is also illustrated in the project. These complementary modalities could help facilitate more extensive lesion analysis and promote additional investigations on multimodal lesion segmentation for MS. Environmentally speaking, training deep learning models can consume a lot of computational resources and electrical energy, particularly for three-dimensional MRI data. To address these needs, the project leveraged the GPUs efficiently and adopted the concept of sliding window inference to

process entire MRI volumes. The trained models can also be used for future inferences, thus avoiding re-training of the models. For future research, the model architecture could be optimized and its computational footprint reduced to further minimize the associated environmental impact of such large-scale medical AI applications.

5.2.3 Ethical Aspects

The ethical issues are particularly relevant in this project, as it relates to medical imaging data. Patient privacy and confidentiality need to be safeguarded during data processing, creation of models, and evaluation. All information that might identify a patient should be kept confidential, and should not be revealed in datasets, file names, experimental results, or system outputs.

In addition, there is a clear separation between the training, validation, and testing data at the patient level in the experimental methodology. This not only minimizes the chances of data leakage but also offers a more accurate evaluation of the model's performance. A held out test set also helps to ensure that the results reported are reflective of the model's performance on unseen patient data.

Another important ethical issue is the potential for segmentation errors. Some lesions may be difficult to detect by the models, especially small lesions and low-contrast lesions or other hard-to-see lesions. Therefore, the developed system is not intended to be a substitute for professional medical advice. It should be regarded as a computer-assisted prediction and should only be used clinically after a thorough review and confirmation by qualified medical personnel.

Transparent reporting is crucial to responsible AI research too. By explicitly specifying the model architecture, the pre-processing steps, the training configuration, and the evaluation metrics, other researchers can get a better picture of the developed approach, reproduce it, and critically evaluate it.

5.2.4 Sustainability Plan

It is important to note that long-term sustainability of the system proposed herein will be ensured with ongoing maintenance of the datasets, improving the performance of the models and an efficient utilization of the computational resources. Future research can involve other MRI datasets and more patient groups to test the model further and to make it more generalized in other groups of patients and other imaging conditions.

The trained models and preprocessing pipeline can be viewed as modular and can be subjected to further change, as better model architectures or preprocessing pipelines become available, without having to redesign the whole system. In future work, research can be carried out on techniques like model compression, pruning and knowledge distillation to decrease the computational requirements and make it more accessible for deployment in platforms with limited hardware resources.

In the future, more in-depth research can be carried out in the aspect of multimodal feature fusion, fine-grained detection of small and low-contrast lesions, and validation with more comprehensive datasets. All such enhancements can contribute to the maintainability of the system in the long term and enhance its robustness, speed of calculation and possibly its usefulness in future medical imaging studies.

5.3 Project Management and Financial Analysis

A structured and cost conscious approach was used to manage the development of the proposed MS lesion segmentation system. The project primarily utilized open-source software frameworks and existing computational resources, which helped minimize development and operational expenses. The major costs were associated with internet connectivity, cloud or computational resources, data handling, documentation, and other project-related activities.

Table 5.1: Estimated Cost and Financial Analysis

Component	Estimated Cost (BDT)
Internet and Cloud/Computational Subscription	2000-3000
Software Licenses (Open-source Software)	0
Data Collection and Related Visits	0
Backup Storage and Miscellaneous Expenses	500-1000
Contingency (10% Buffer)	500
Total Estimated Cost	3000-4000

5. 4 Complex Engineering Problem

The development of the proposed automated Multiple Sclerosis (MS) lesion segmentation system involved several complex engineering problems requiring advanced knowledge of deep learning, medical image processing, three-dimensional data analysis, and experimental evaluation. The project required the integration of MRI preprocessing, multimodal feature extraction, model training, three-dimensional inference, and quantitative evaluation within a single workflow.

5.4.1 Complex Engineering Problem solving

The major engineering challenges and their relationship to complex engineering problem-solving characteristics are discussed below.

Table 5.2: Mapping with Complex Engineering Problem.

EP1 Dept of Knowled ge	EP2 Range Of Conflicting Requireme nts	EP3 Depth of Analys is	EP4 Familiari ty of Issues	EP5 Extent of Applica ble Codes	EP6 Extent Of Stake- holder Involveme nt	EP7 Interdepende nce
✓		✓	✓			✓

Justification of EP1 attainment:

The project applied advanced knowledge of deep learning, medical image processing, three-dimensional convolutional neural networks, MRI preprocessing, and segmentation evaluation. Multiple architectures, including 3D U-Net, 3D Attention U-Net, UNETR, and the proposed gated residual multimodal model, were developed and evaluated for MS lesion segmentation.

Justification of EP3 attainment:

Systematic experimentation was conducted by varying and evaluating different model architectures, preprocessing procedures, training configurations, and inference strategies. The performance of the developed models was analyzed using Dice coefficient, precision, recall, and Intersection over Union (IoU) at both global and patient-wise levels.

Justification of EP4 attainment:

The project addressed several technically challenging and relatively specialized issues, including three-dimensional MRI processing, N4 bias-field correction, non-zero voxel normalization, GPU memory limitations, patient-level data separation, and segmentation of lesions with different sizes and intensity characteristics. The proposed multimodal model further required appropriate integration of FLAIR, T1-weighted, and T2-weighted information.

Justification of EP7 attainment:

The project required the integration of several interdependent stages, including MRI data preparation, preprocessing, model training, checkpoint management, sliding-window inference, prediction generation, and quantitative evaluation. A change in one stage could affect subsequent stages; therefore, the complete workflow had to be designed and maintained as a coherent pipeline.

Mapping with Knowledge Profile for EP1

Table 5.3: Mapping with knowledge Profile.

K1 Natu ral Scien ce	K2 Mathem atics	K3 Engineeri ng Fundame ntals	K4 Special ist Knowle dge	K5 Enginee ring Design	K6 Enginee ring Practice	K7 Comprehe nsion	K8 Resear ch Literat ure
		✓	✓	✓	✓		✓

5.4.2 Engineering Activities

In terms of engineering activities, the project involved several activities associated with complex engineering practice, particularly in resource utilization, system interaction, innovation, societal impact, and technical familiarity. Mapping with Complex Engineering Activities

Table 5.4: Mapping with Complex Engineering Activities.

EA1 Range of re- sources	EA2 Level of Interaction	EA3 Innovation	EA4 Consequences for society and environment	EA5 Familiarity
✓			✓	✓

Justification of EA1 attainment:

The project required a range of computational and software resources, including the MSLesSeg MRI dataset, GPU-based computation, Python, PyTorch, MONAI, SimpleITK, NiBabel, NumPy, and other open-source libraries. Different resources were integrated for MRI preprocessing, model development, training, 3D inference, and quantitative evaluation. Furthermore, various segmentation architectures such as 3D U-Net, Attention U-Net, UNETR and the proposed gated residual multimodal model were also developed and tested.

Justification of EA4 attainment:

Though the developed system could not be used as a clinical diagnostic tool, automated MS lesion segmentation can have a positive impact on society by saving the time spent on manual segmentation and aiding medical image analysis and research. GPU-based processing and a structured 3D inference strategy also allow for a practically implementable use. Meanwhile, the limitations and segmentation errors of the system should be taken into account before the clinical application.

Justification of EA5 attainment:

The project involved the use of existing knowledge in Python programming, machine learning, deep learning, medical image processing, and MRI analysis. These technical foundations facilitated the preprocessing, model training, checkpoint handling, 3D sliding window inference and quantitative evaluation. Additionally, the project involved the learning and application of specific concepts like N4 bias correction, multimodal feature fusion, residual correction and lesion-level analysis.

5.5 Summary

This chapter addressed the project's compliance with engineering standards, ethics, social responsibility, and sustainable development. To ensure reliability, reproducibility, and maintainability in the MS lesion segmentation workflow, the system was designed with well-known software, hardware, and communication protocols. The ethical concerns, such as patient privacy, proper management of medical data, and clear reporting of model limitations, were also taken into account.

The chapter also explored the potential of automated MS lesion segmentation and its social and environmental impact. The system can specifically lower the labour for lesion analysis and facilitate further studies in medical image processing. The project was also viewed from a complex problem solving and engineering activities point of view, highlighting the integration of medical image processing, deep learning, multi modal MRI analysis, GPU based computation and quantitative performance evaluation.

Lastly, project management and financial aspects were discussed to determine the feasibility of developing and improving the proposed system. All these considerations suggest that the system can be developed and further improved within reasonable computational and economic limits.

Chapter 6

Conclusion

The chapter reflects the general conclusion of the proposed Multiple Sclerosis (MS) lesion segmentation project that should be automated. It summarizes the major accomplishments of the research, comments on the limitations faced in the process of developing and testing the system and outlines the potential future work directions. The project was aimed at using deep learning and medical image processing methods to automatically segment MS lesions on the three-dimensional MRI data. Special attention was paid to better lesion detection and the possibility of using complementary information of several MRI modalities. The results, which are shown in the course of the research, offer a premise to evaluate the efficiency of the created methods and see the ways of further enhancement.

6.1 Summary

This study suggested an automated deep learning method of Multiple Sclerosis (MS) lesion segmentation with three-dimensional MRI data. The main aim was to design and test a sound segmentation scheme to detect MS lesions on multimodal MRI scans and to overcome issues of lesion size, intensity change and the complexity of volumetric medical images. The research was conducted using a systematic approach that included the preparation of the dataset, MRI preprocessing, model construction, training, three-dimensional sliding-window inference, and quantitative analysis. The dataset was MSLesSeg, which included FLAIR, T1-weighted, and T2-weighted MRI modalities and lesion masks. The preprocessing pipeline involved the N4 bias-field correction and non-zero voxel Z-score normalization. The training and testing sets were separated at the patient level to reduce the chances of data leakage and guarantee independent assessment of the model performance.

A number of deep learning models were tested and trained, such as 3D U-Net, 3D Attention U-Net, and UNETR, and FLAIR MRI was utilized as the main input. Moreover, a suggested gated residual multimodal model was created based on FLAIR, T1-weighted, and T2-weighted MRI data. The architecture proposed included gated feature fusion and bounded residual correction to explore the possibility of complementary information between various MRI modalities to enhance MS lesion segmentation. The Dice coefficient, precision, recall, and Intersection over Union (IoU) were used to evaluate model performance, as well as patient-wise and lesion-level performance. UNETR was the best of the models tested with a global Dice score of 0.7123, global precision of 0.7472 and global IoU of 0.5531. The proposed Phase 2 model had a global Dice score of 0.7009 and the highest global recall of 0.7406, which means that it is more sensitive to lesion regions. The lesion-level comparison between UNETR and Phase 2 also revealed an increase in the mean lesion recall between 0.4200 and 0.4830. Moreover, the rate of lesion detection was higher with Phase 2 (0.7162) compared to UNETR (0.6613).

In general, this study illustrates the use of various three-dimensional deep learning models and multimodal MRI processing in a well-organized MS lesion segmentation pipeline. The results also point to the difficulties in identifying small and low-intensity lesions, which are also significant areas to be improved. The generated methodology offers

a basis of future studies to enhance the strength, external validity, and possible clinical utility of automated MS lesion segmentation systems.

6.2 Limitation

The project successfully accomplished its primary goals, but there were some limitations found while developing and testing the proposed MS lesion segmentation system. The experimental results should be interpreted within these constraints and potential avenues for future improvement should be considered. One of the main limitations was the relatively limited size of the dataset. While the MSLesSeg dataset offers well-prepared 3D MRI scans and their corresponding lesion labels, it has a limited number of patients and scans for training deep learning models. This could impact the generalizability of the trained models to other patient populations, scanners, acquisition protocols, or clinical settings with MRI data.

The other drawback was the computational complexity of 3D image segmentation of medical images. For architecture like UNETR, a lot of GPU memory and time is needed to process volumetric MRI data. Consequently, the experiments were performed in the hardware available, and larger model configurations, more extensive hyperparameter optimization and other more intensive experiments were not explored. One of the other issues the models had was identifying small, faint lesions. Lesion level analysis revealed a difference in segmentation performance based on lesion size and FLAIR intensity. The smaller and dimmer lesions tended to be less detectable than larger and brighter lesions, and thus resulted in missed lesions and lower lesion-level recall in these cases. The proposed multimodal model also failed to score the best overall Dice score even though it scored the best global recall. The gated feature fusion and the residual correction were added to the architecture, which added to the complexity of the architecture, as FLAIR, T1-weighted, and T2-weighted information were incorporated. The results indicate that a combination of multiple MRI modalities does not always lead to an improvement in overall segmentation overlap and the success of multimodal information relies on the way that the available features are combined.

Lastly, the analysis was done with the available held-out MSLesSeg test set and the results are yet to be confirmed with independent external data. The differences in MRI scanners, acquisition protocols, patient populations, and lesion characteristics could influence the model performance when used on other datasets. Thus, the system still needs to be validated with larger, independent, and hopefully multi-center datasets before it can be deemed suitable to be used in practice.

6.3 Future Work

On the basis of the results and shortcomings of the present research, one can think of a number of directions to be followed in the further development of the proposed MS lesion segmentation system. Among the key concerns, the expansion of the dataset in terms of size and diversity should be prioritized, incorporating MRI scans of larger and more diverse populations of patients, imaging centers, scanners, and acquisition protocols. A more varied dataset would be useful in enhancing the strength and generalizability of the segmentation models as well as offer a better foundation to independent external validation.

As more computational resources become available, future research might consider more complex and larger three-dimensional segmentation architectures and perform more comprehensive hyperparameter optimization. Further development of the proposed multimodal approach could also focus on more effective ways of combining FLAIR, T1-weighted, and T2-weighted information. Special consideration should be paid to small and low-intensity lesions, which were still difficult to detect in the present experiments.

The next research might also consider other preprocessing and data augmentation methods to enhance the resilience of the models to changes in MRI intensity and acquisition parameters. Lesion-level analysis would be more detailed to evaluate the performance of detection by factors like lesion size, intensity, and anatomical location.

The future work will also focus on clinical and external validation. The models developed are to be tested on independent datasets to find out whether they can be used with various patient populations and imaging settings. Cooperation with medical workers might also be used to determine the clinical significance of the segmentation findings and detect failure cases that cannot be detected only using quantitative indicators.

Lastly, optimization methods like knowledge distillation, pruning, and quantization could be considered to further lower computational needs and make the system more appropriate for deployment. A user-friendly interface could also be created to visualize the MRI scans, the ground truth and predicted lesion masks, as well as the quantitative evaluation results. These enhancements, combined, may contribute to the development of a more robust, scalable, and accessible computer-aided system for MS lesion analysis with appropriate clinical oversight, from the current research prototype.

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